

DPPL-xx

SOFTWARE DESIGN DESCRIPTION

Sentinel: A Geopolitical Travel Safety Advisor Website

for:

Travelers Worldwide

Prepared by:



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		<i>DPPL-xx</i>		<i>1 / 99</i>
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LIST OF CHANGES

Revision	Description
A	Several updates were made to this document. First, the names of Functional Requirements (FRs) and Use Cases are updated, whereas the Use Cases now use active verbs to make them different from the Functional Requirements (FRs). These changes are reflected in the traceability matrix. Then, Use case scenarios are rewritten from the actors' point of view. The Class Diagram is updated to follow the MVC design pattern, and the main system algorithms and SQL queries are improved. For better organization, a list of figures, a list of tables, and a list of changed pages are included. Finally, Attachment C was added to provide the Mermaid code for the MVC Class diagram.

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1. Introduction

1.1 Document Purpose

This document defines the scope, structure, and technical blueprint of the Sentinel Travel Safety Advisory platform. Its main purpose is to act as a clear and standardized reference that guides the planning, design, and implementation of the Sentinel system.

The document explains the system's objectives, important definitions and terminology, and the functional requirements of Sentinel. It also describes the overall system design, including hardware and software requirements, deployment environment, and external integrations. In addition, it defines the software architecture, system modules, class structures, and data handling approach. The document further presents use case realizations, user interface designs, sequence diagrams, algorithms, and database query designs, and establishes traceability between functional requirements and use cases to ensure completeness and consistency.

This document is intended for project stakeholders and evaluators to understand the system's goals, scope, and feasibility. It is also meant for system analysts and designers to support architectural and design decisions, for developers as a technical reference during implementation, for testers and quality assurance teams to verify system behavior against defined requirements, and for academic reviewers and instructors for evaluation and documentation verification.

Overall, this DPPL document serves as a single authoritative source that ensures all team members and stakeholders have a shared and consistent understanding of the Sentinel system's design, functionality, and implementation plan.

1.2 Document Scope

This document serves as the technical blueprint for Sentinel, a specialized web application designed to help travelers and travel agencies navigate global geopolitical risks. It defines the system architecture, functional modules, and data structures required to build the application.

The scope covers several core components, beginning with the system architecture which provides design specifications for the functional modules, including role-based access control, the Web UI, and external API integrations for government and news data. It further includes functional designs and use case realizations for the Interactive Risk Map, destination-specific Safety Alerts and Emergency Contacts , and Destination Information retrieval.

Additionally, the document outlines the technical workflow for the Data Verification Pipeline. This module allows Affiliated Journalists to submit on-ground intel which is then vetted for accuracy by Risk Experts. For B2B capabilities, the scope details the Travel Route Assistant feature, which is specifically designed for Travel Agency Partners to assess and mitigate safety risks along planned itineraries. Data modeling is addressed through structural designs including the Entity Relationship Diagram (ERD), Relational Schema, and SQL DDL query designs used for managing user roles, risk metrics, and destination profiles.

Finally, the system boundaries establish that Sentinel operates exclusively as an informational and decision-support tool for safety intelligence and route recommendations. This design does not extend to direct flight or hotel booking services, financial transaction processing, or physical emergency response and extraction services.

1.3 Terms and Definitions

The following table defines key terms used throughout this document to ensure clarity and consistent understanding of system concepts.

Term	Definition
Sentinel	“Sentinel” is an intelligent travel-safety advisory platform that centralizes verified geopolitical data, real-time alerts, and official travel advisories to help users make informed and safer travel decisions for any chosen destination.
Safety Alert	A real-time notification displayed on the platform when urgent issues arise, such as disasters, conflict escalation, or emergencies at a selected destination.
Geopolitical Risk	Any threat originating from political instability, conflict, terrorism, civil unrest, or other factors that may endanger travelers.
Interactive Risk Map	A visual map in Sentinel that uses color-coded shading to display the geopolitical risk level of countries or regions.
Verification Workflow	The process where journalists submit data, risk experts verify or reject it, and verified information is published on the platform.
Reminder	A saved safety tip or key point that a traveler stores for future use.
Destination Information	The combined safety data, alerts, risk levels, cultural notes, and emergency contacts displayed when a user selects a destination.
Search Bar	A feature allowing users to search for destination information quickly.
Optimal Route	A suggested travel route based on real-time safety and geopolitical conditions, available to Travel Agency Partners.

Table 1. Terms and definitions

1.4 References

- [1] T. S. Gov, “Travel Advisories | Travel.State.gov,” State.gov, 2025. Available: <https://travel.state.gov/en/international-travel/travel-advisories.html>
- [2] World Travel Guide, “World Travel Guide,” World Travel Guide, 2019. Available: <https://www.worldtravelguide.net/>
- [3] CnTraveler, “Condé Nast Traveler - Travel Reviews, News, Guides and Tips,” Condé Nast Traveler, 2019. Available: <https://www.cntraveler.com/>

2 Global Design

2.1 Implementation Environment Plan

2.1.1 Hardware Requirements

Sentinel is an informational resource, meaning that the hardware focus is ensuring that the server can handle data parsing from external sources. Such hardware is detailed in the following table.

Components	Recommended Specifications
Processor	4 Core Xeon/EPYC (Production Server)
RAM	8 GB (To handle data caching/concurrency)
Storage	100 GB++ SSD (For logging and data archives)
Connectivity	100 Mbps+ (For real-time advisory fetching)

Table 2. Hardware components

2.1.2 Software Requirement

These are the tools the team will use to make Sentinel software work and publish it to the international market. The required software is listed in the following table.

Components	Description/Version
Operating System	Linux (Ubuntu 22.04 LTS) or Windows 11
Programming Language	Python 3.11 (Ideal for data scraping/APIs)
Web Framework	Django (For rapid data delivery)
Database	PostgreSQL (Relational data for country profiles and good security)
Caching	Redis (To store frequent travel alerts)
Version Control	Git and Github

Table 3. Software components

2.1.3 Deployment and External Environment

As Sentinel relies on external information or public data, this section defines the application's hosting environment and its integration with public data sources. The following table lists the detailed features used for supporting the system.

Feature	Specifications
Cloud Hosting	AWS (EC2/S3)
Data Sources	API connections to Travel sites, Gov sites, and sources from experts
Security	SSL/TLS Encryption (HTTPS) for user confidence

Table 4. Features

2.2 Software Architecture Description

Sentinel's architecture is built upon specific modules that facilitate core system functionality. The individual components of these modules are detailed in the following table.

No.	Component Name	Description
1	Web User Interface Module	Provides a web interface for Travelers, Travel Agency Partners, Journalists, Risk Experts, and Administrators.
2	Authentication and Authorization Module	Handles user authentication and role-based access management, encompassing login and registration modules (FR-08, FR-09, FR-15).
3	Destination Information Module	Manages and visualizes destination data, including risk assessments and geopolitical status updates (FR-01, FR-03, FR-04).
4	Safety Alerts and Emergency Module	Facilitates location-based safety alerts and emergency contact retrieval (FR-02).
5	Interactive Risk Map Module	Features an interactive map providing spatial visualization of geopolitical risk levels (FR-07).
6	Search, Filter, and Sorting Module	Manages destination search functionality, including data filtering and sorting based on risk indices (FR-10, FR-12).
7	Reminder Management Module	Manages data persistence and retrieval for user-defined safety reminders (FR-05).

8	Risk Report Verification Workflow Module	Facilitates the data verification workflow between Journalists and Data Analysts/Risk Experts (FR-11).
9	Route Planning Module	Provides optimal route recommendations based on real-time geopolitical conditions and risk analysis (FR-14).
10	Content and Data Management Module	Enables Administrative users to oversee system data management and content governance (FR-13).
11	External API Integration Module	Integrates the system with mapping APIs, government advisories, and external news sources.
12	Database Management Module	Manages data persistence for user profiles, destination catalogs, risk metrics, and verification records.

Table 5. Module components

3 Detailed Design

3.1 Use Case Realization

The realization of each use case is defined in the following table.

No.	Use Case Name	Use Case Description
1.	View Destination Profile	The system displays verified and up-to-date safety information upon the user's selection of a destination, including the calculated risk level and geopolitical status.
2.	Check Alerts and Contacts	When a destination is selected, the system displays recent safety alerts from trusted sources and the relevant emergency contact numbers.
3.	Review Key Safety Points	The system highlights summarized key safety points and actionable advice derived from verified intelligence .
4.	View Short Safety Tips	The system displays short safety tips, cultural notes, and awareness messages when a traveler selects a destination.
5.	Save Safety Reminder	The system allows the traveler to save important safety reminders for future access.
6.	View Short Safety Quote	The system displays a short, awareness-promoting safety quote when a user first opens the website.
7.	Explore Interactive Risk Map	The system shows an interactive map where geopolitical risk areas are differentiated by color shading.
8.	Register New Account	The system allows a new user to register for a separate account as either a "Traveler" or a "Travel Agency Partner."
9.	Manage User Profiles	This use case allows the system administrator to create, update, and delete user accounts to control access, manage credentials, and ensure accurate user information.
10.	Search for Destinations	This use case enables users to search for their desired destinations using a search bar that provides relevant and quick results.
11.	Verify Submitted Data	The system shall enable an Affiliated Journalist to submit information for verification to a Risk Expert. The system shall notify the Risk Expert of the pending verification request. The system shall allow the Risk

		Expert to confirm or reject the geopolitical data accuracy.
12.	Filter and Sort Results	The system should allow users to filter and sort destinations or news articles based on risk levels, such as low, medium, or high.
13.	Administer System Data	Administrators have the authority to manage database content, including adding, updating, and deleting data related to users, destinations, and safety information
14.	Generate Travel Route	Travel agencies will have access to an intelligent route-planning feature that suggests optimal routes and destinations based on geopolitical conditions and safety data.
15.	Log into Application	The system shall provide separate login and registration functionality for the Traveler role. The system shall provide separate login and registration functionality for the Travel Agency Partner role.

Table 6. Use cases and descriptions

3.1.1 Use Case #1 Search Destination Safety Profile

3.1.1.1 Use Case Scenario #1 Search Destination Safety Profile

Pre-Condition: Actor has access to the Sentinel platform and is viewing the homepage or interactive map.

Description: Actor searches for a specific destination to view its comprehensive safety profile, including risk level, geopolitical status, and verified safety information.

	Main Flow
1.	Actor enters a destination name in the search bar or clicks on a location on the interactive map.
2.	Actor submits the search query.
3.	Actor receives search results showing matching destinations.
4.	Actor selects their desired destination from the results.
5.	Actor views the destination profile page, and the system shows options for alerts and advisories.
	Alternative Flow
1.	Actor enters a destination name in the search bar.

2.	Actor submits the search query.
3.	Actor sees a "No results found" message.
4.	Actor views suggested similar regions or nearby destinations.
5.	Actor either selects one of the suggested alternatives or modifies their search query and searches again or returns to the map to browse visually.

Table 7. Use case scenario #1 flow

Post-Condition: Actor has accessed destination safety information or received guidance on alternative search options when results are unavailable.

3.1.1.2 Wireframe #1 Search Destination Safety Profile

The following figure illustrates the wireframe.

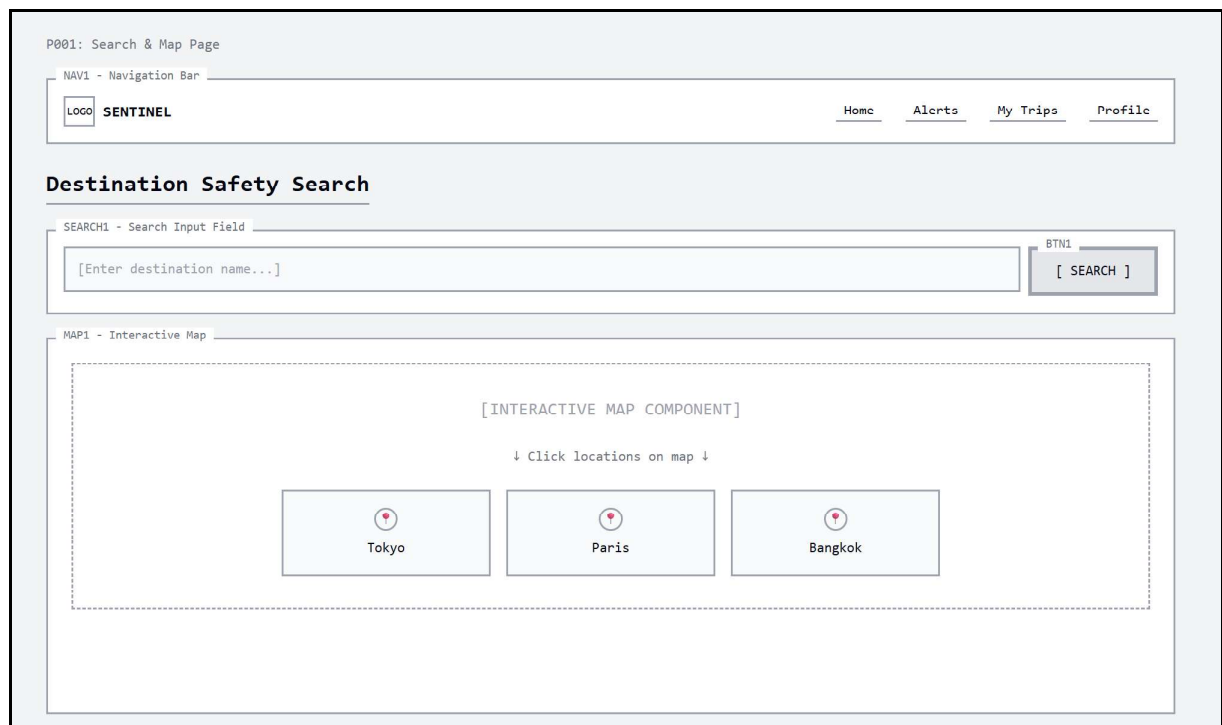


Figure 1. Use case #1 wireframe A

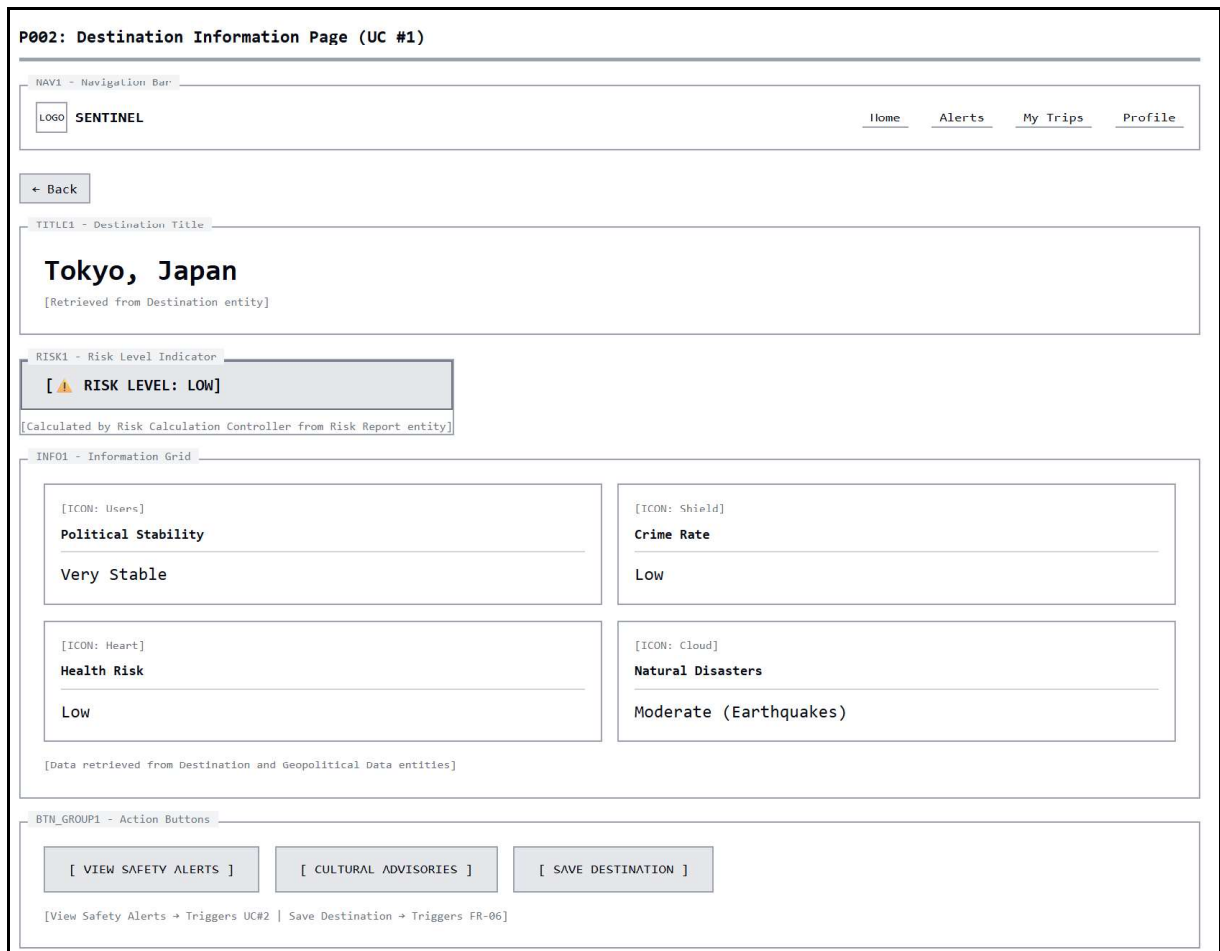


Figure 2. Use case #1 wireframe B

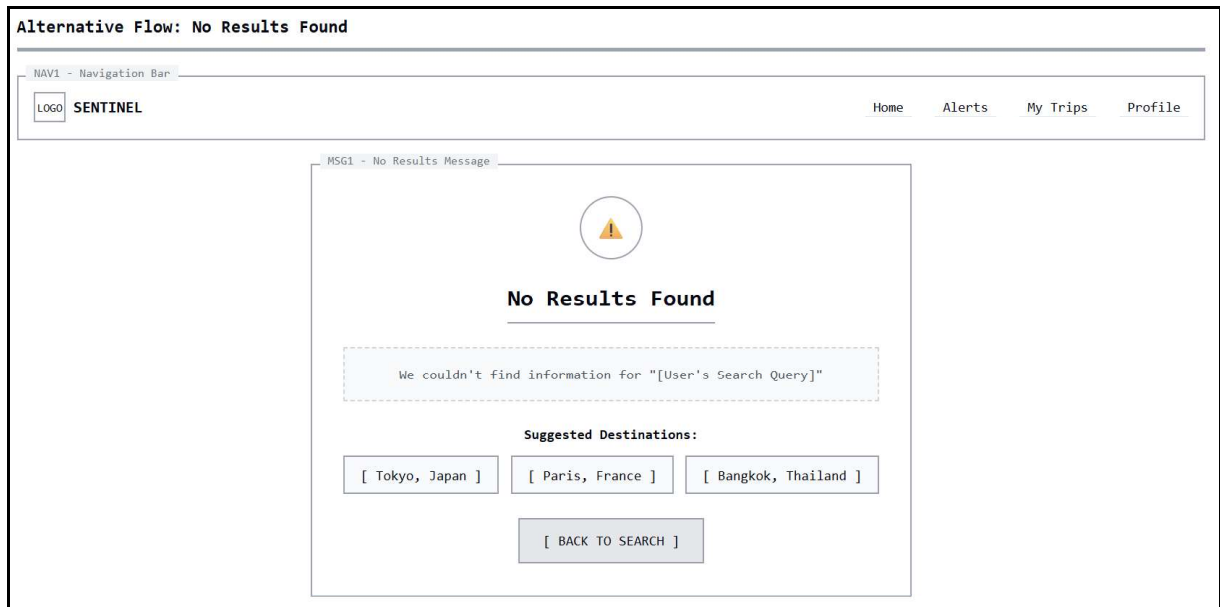


Figure 3. Use case #1 wireframe C

3.1.1.3 UI Design and UI Description Object #1 Search Destination Safety Profile

UI Description Design Table

ID PAGE	PAGE NAME	DESCRIPTION
P001	Search & Map Page	Landing page with search bar and interactive map for destination selection.
P002	Destination Information Page	Displays destination details with risk level and geopolitical status.

Table 8. Use case #1 UI Description Design Table

Interface P001: Search & Map Page

Object ID	TYPE	LABEL	DESCRIPTION
NAV1	Navigation Bar	Navigation Menu	Displays main menu options. When clicked, navigates to respective pages
SEARCH1	Text Input	Search Input Field	Allows actor to enter destination name. Triggers search functionality
BTN1	Button	Search Button	When clicked, activates search function and queries database using geopolitical identifier
MAP1	Interactive Component	Interactive Map	Interactive map component. When actor clicks on country/region, triggers View Destination Information use case

Table 9. Use case #1 interface details

Interface P002: Destination Information Page (UC #1)

Object Id	TYPE	LABEL	DESCRIPTION
NAV1	Navigation Bar	Navigation Menu	Displays main menu options for

			navigation
TITLE1	Text Label	Destination Title	Displays selected destination name (retrieved from Destination entity)
RISK1	Status Badge	Risk Level Indicator	Displays calculated risk level (Low/Moderate/High/Critical) - System processes Risk Report entity
INFO1	Data Grid	Information Grid	Displays 4 information boxes: Political Stability, Crime Rate, Health Risk, Natural Disasters (retrieved from Destination and Geopolitical Data entities)
BTN GROUP1	Button Group	Action Buttons	Contains 3 buttons: "View Safety Alerts" (triggers UC#2), "Cultural Advisories", "Save Destination" (triggers FR-06)
MSG1	Alert Message	No Results Message	Alternative Flow - Displayed when search query yields zero matching destinations. Shows suggestions for similar regions

Table 10. Use case #1 interface details

3.1.1.4 Object Identification and Class Type #1 Search Destination Safety Profile

DESIGN OBJECT TABLE

No.	New Object Name	Class Type
1	Search Interface	Boundary (UI Component)
2	Map Interface	Boundary (Interactive UI Component)
3	Destination Controller	Controller (Processing Logic)
4	Destination	Entity (Database Table)
5	Risk Report	Entity (Database Table)
6	Geopolitical Data	Entity (Database Table)
7	Destination Display Interface	Boundary (Result Display UI)
8	Risk Calculation Controller	Controller (Risk Level Calculation)
9	Search Query Validator	Controller (Input Validation)
10	No Results Interface	Boundary (Alternative Flow UI)

Table 11. Use case #1 Design Object Table

3.1.1.5 Sequence Diagram #1 Search Destination Safety Profile

The following sequence diagram outlines the chronological flow of messages for this use case shown in the figure below.

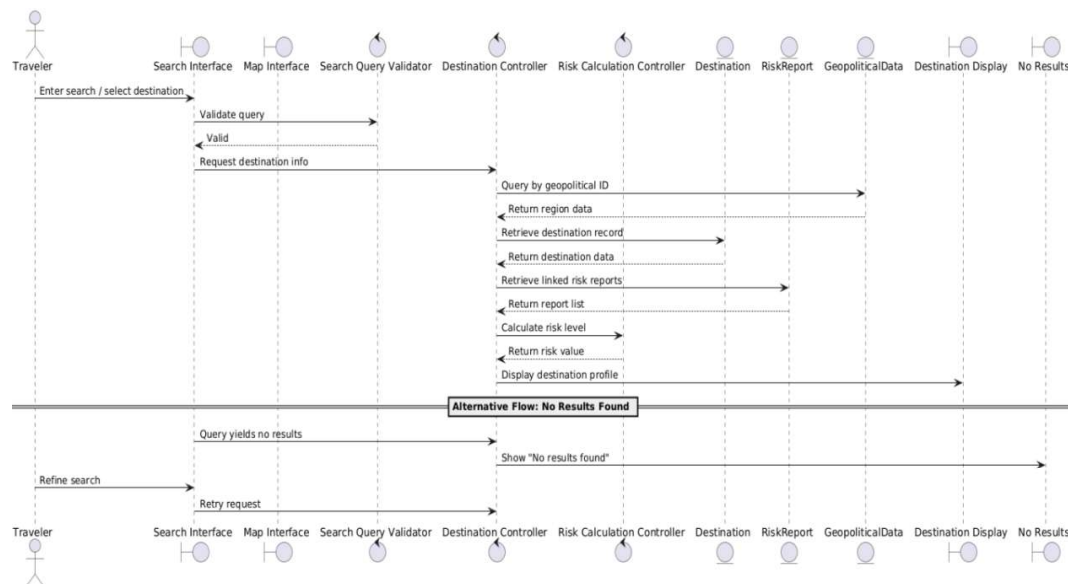


Figure 4. Use case #1 sequence diagram

3.1.2 Use Case #2 Check Real-time Destination Safety Alerts

3.1.2.1 Use Case Scenario #2 Check Real-time Destination Safety Alerts

Pre-Condition: Actor is logged into the system and has selected a destination (from UC #1).

Description: Actor accesses real-time safety alerts and emergency contact information for their selected destination to stay informed about current risks.

	Main Flow
1.	Actor navigates to the selected destination's detail page.
2.	Actor scrolls to or clicks on the "Safety Alerts" section.
3.	Actor reviews the displayed recent safety alerts, including alert title and severity level ,description of the situation and date and source of the alert.
4.	Actor scrolls to or clicks on the "Emergency Contacts" section.
5.	Actor views verified emergency contact numbers
	Alternative Flow
1.	Actor navigates to the selected destination's detail page.
2.	Actor scrolls to or clicks on the "Safety Alerts" section.
3.	Actor sees a message: "No current alerts for this destination."
4.	Actor views suggested similar regions or nearby destinations.
5.	Actor either selects one of the suggested alternatives or modifies their search query and searches again or returns to the map to browse visually.

Table 12. Use case scenario #2 flow

Post-Condition: Actor is informed of current safety alerts (or their absence), has emergency contact information, and has optionally saved or shared this information for future reference.

3.1.2.2 Wireframe #2 Check Real-time Destination Safety Alerts

The following figure is the wireframe.

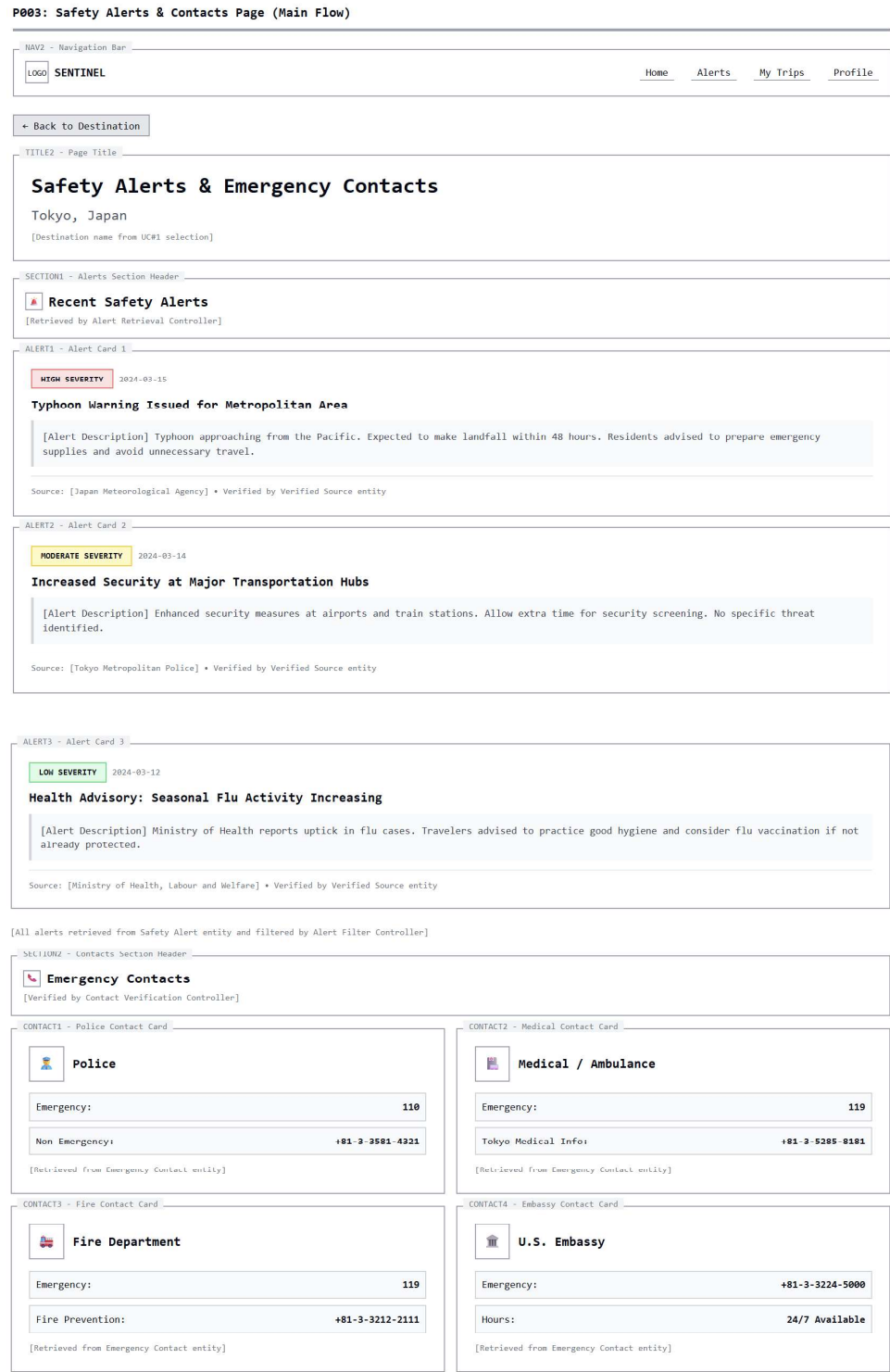


Figure 5. Use case #2 wireframe

3.1.2.3 UI Design and UI Description Object #2 Check Real-time Destination Safety Alert

UI Object Description Table

SCREEN ID	SCREEN NAME	DESCRIPTION
P003	Safety Alerts & Contacts Page	Shows recent safety alerts and emergency contact information.

Table 13. Use case #2 UI Object Description Table

Interface P003: Safety Alerts & Contacts Page

Object Id	TYPE	LABEL	DESCRIPTION
NAV2	Navigation Bar	Navigation Menu	Displays main navigation menu
TITLE2	Text Label	Page Title	Displays page title with destination name
SECTION1	Section Header	Alerts Section Header	Header for Recent Safety Alerts section
ALERT1	Card Component	Alert Card 1	Displays first safety alert with title, description, date, and source (retrieved from Safety Alert entity)
ALERT2	Card Component	Alert Card 2	Displays second safety alert with details
ALERT3	Card Component	Alert Card 3	Displays third safety alert with details
SECTION2	Section Header	Contacts Section Header	Header for Emergency Contacts section
CONTACT1	Card Component	Police Contact Card	Displays police emergency contact information (retrieved from

			Emergency Contact entity)
CONTACT2	Card Component	Medical Contact Card	Displays medical/ambulance emergency contacts
CONTACT3	Card Component	Fire Contact Card	Displays fire department emergency contacts
CONTACT4	Card Component	Embassy Contact Card	Displays embassy/consulate contact information
MSG2	Alert Message	No Alerts Message	Alternative Flow - Displayed when no recent alerts available. Shows general emergency information message

Table 14. Use case #2 interface details

3.1.2.4 Identification of New Objects and Class Type #2 Check Real-time Destination Safety Alert

DESIGN OBJECT TABLE

No.	New Object Name	Class Type
1	Alerts Page Interface	Boundary (Main UI Page)
2	Alert Retrieval Controller	Controller (Data Fetching Logic)
3	Safety Alert	Entity (Database Table)
4	Emergency Contact	Entity (Database Table)
5	Verified Source	Entity (Database Table)
6	Alert Display Component	Boundary (Alert Card UI)
7	Contact Display Component	Boundary (Contact Card UI)
8	Alert Filter Controller	Controller (Filter Recent Alerts)

9	Contact Verification Controller	Controller (Verify Contact Accuracy)
10	No Alerts Interface	Boundary (Alternative Flow UI)

Table 15. Use case #2 Design Object Table

3.1.2.5 Sequence Diagram #2 Check Real-time Destination Safety Alerts

The following sequence diagram outlines the chronological flow of messages for this use case shown in the figure below.

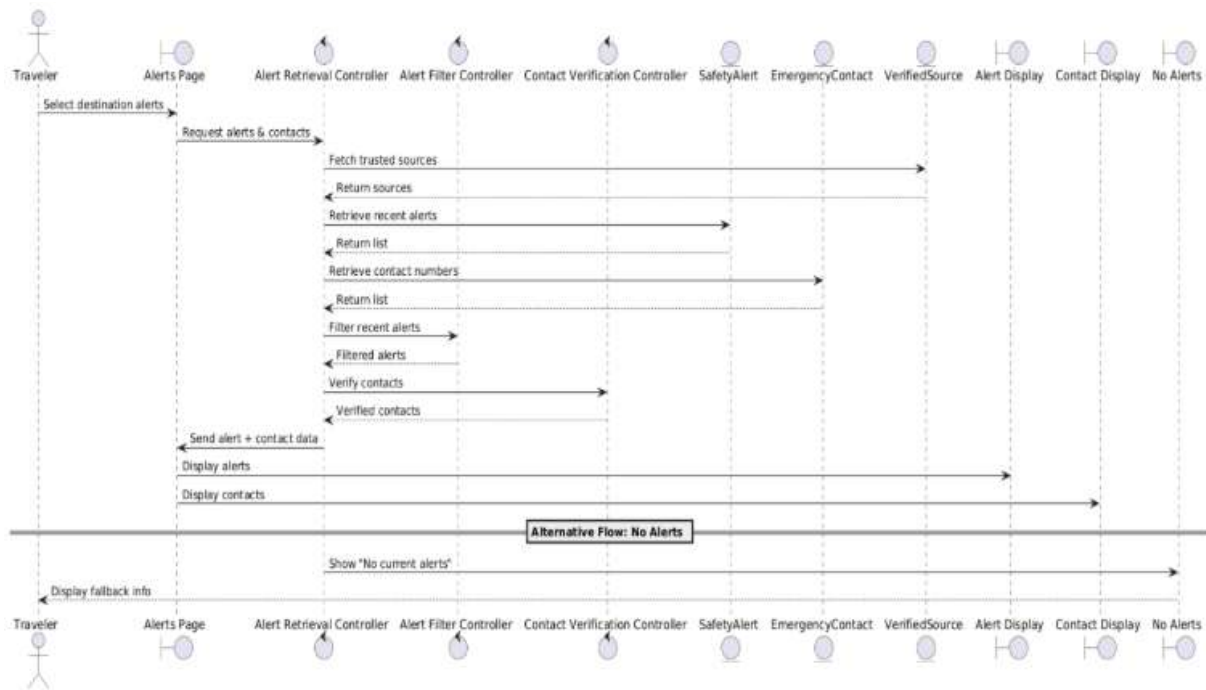


Figure 6. Use case #3 sequence diagram

3.1.3 Use Case #3 Review Key Safety Points

3.1.3.1 Use Case Scenario #3 Review Key Safety Points

Pre-Condition: Actor is viewing a destination's information page (after completing UC #1) and key safety points have been generated for this destination.

Description: Actor reviews summarized, actionable safety advice specific to their destination and saves important points for later reference.

	Main Flow
1.	Actor scrolls through the Destination Information page.
2.	Actor locates the "Key Safety Points" section (displayed with highlighting/distinct

	design).
3.	Actor reads through multiple safety points, each containing risk description, actionable advice, source verification indicator
4.	Actor identifies a particularly relevant safety point.
5.	Actor clicks the "Save Reminder" button next to that point.
6.	Actor sees confirmation that the reminder has been saved to their account.
7.	Actor repeats steps 4-6 for additional safety points as needed.
	Alternative Flow
1.	Actor scrolls through the destination information page.
2.	Actor reaches the "Key Safety Points" section.
3.	Actor sees a message: "Detailed safety highlights are being compiled. General safety information is available above."
4.	Actor views the general fallback information.

Table 16. Use case scenario #3 flow

Post-Condition: Actor has reviewed and optionally saved key safety points, or has accessed alternative safety information when detailed highlights are unavailable, ensuring they have actionable safety knowledge for trip planning.

3.1.3.2 Wireframe #3 Review Key Safety Points

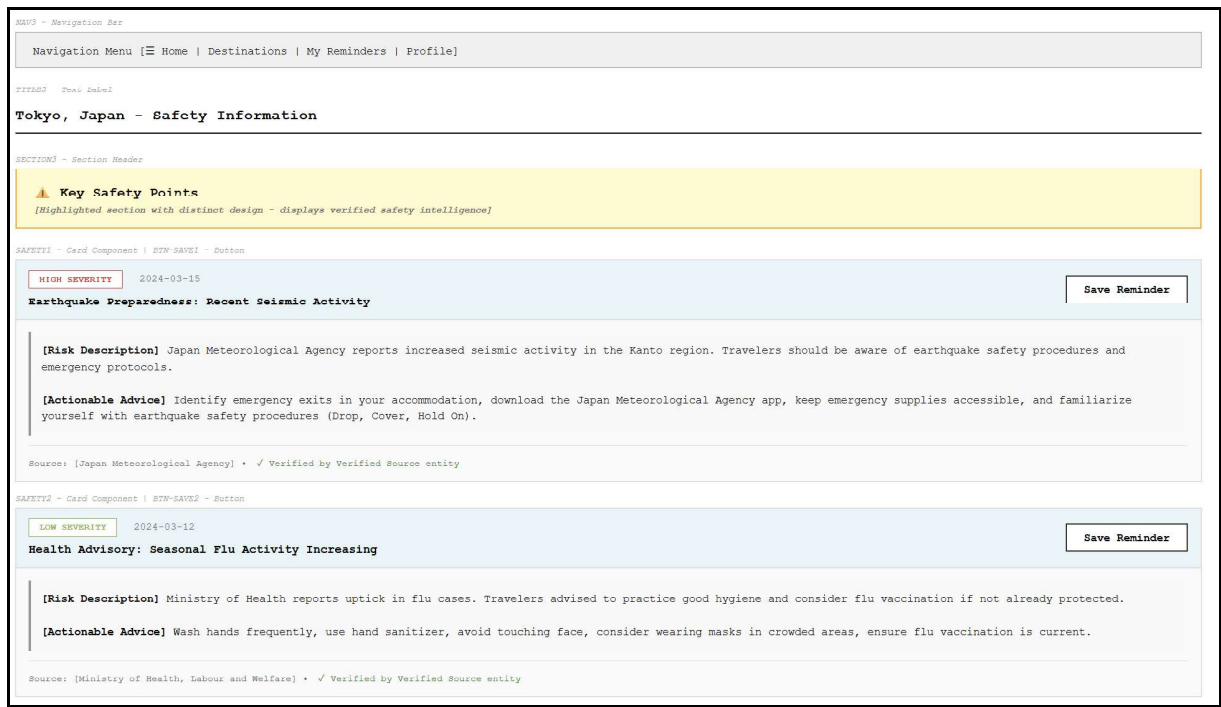


Figure 7. Use case #3 wireframe

3.1.3.3 UI Design and UI Description Object #3 <Review Key Safety Points>

UI Object Description Table

SCREEN ID	SCREEN NAME	DESCRIPTION
P004	Key Safety Points Page	Displays highlighted key safety points with save functionality.

Table 17. Use case #3 UI Object Description Table

Interface P003: Key Safety Points Page

Object Id	TYPE	LABEL	DESCRIPTION
NAV3	Navigation Bar	Navigation Menu	Displays main navigation menu
TITLE3	Text Label	Page Title	Displays page title with destination name

SECTION3	Section Header	Key Safety Points Section	Header for highlighted key safety points section (distinct/highlighted design)
SAFETY1	Card Component	Safety Point Card 1	Displays first key safety point with risk description and actionable advice
BTN-SAVE1	Button	Save Button 1	When clicked, saves safety point as reminder (triggers FR-06, stores in User Reminder entity)
SAFETY2	Card Component	Safety Point Card 2	Displays second key safety point with details
BTN-SAVE2	Button	Save Button 2	Saves second safety point as reminder
SAFETY3	Card Component	Safety Point Card 3	Displays third key safety point with details
BTN-SAVE3	Button	Save Button 3	Saves third safety point as reminder
SAFETY4	Card Component	Safety Point Card 4	Displays fourth key safety point with details
BTN-SAVE4	Button	Save Button 4	Saves fourth safety point as reminder

SAFETY5	Card Component	Safety Point Card 5	Displays fifth key safety point with details
BTN-SAVE5	Button	Save Button 5	Saves fifth safety point as reminder
MSG3	Alert Message		

Table 18. Use case #3 interface details

3.1.3.4 Identification of New Objects and Class Type #3 Review Key Safety Points

DESIGN OBJECT TABLE

No.	New Object Name	Class Type
1	Safety Points Interface	Boundary (Main UI Page)
2	Safety Point Controller	Controller (Processing Logic)
3	Key Safety Point	Entity (Database Table)
4	Destination	Entity (Database Table - Linked)
5	Verified Intelligence	Entity (Database Table - FR-12 Source)
6	Safety Point Display Component	Boundary (Safety Card UI Component)
7	Reminder Controller	Controller (Save Reminder - FR-06)
8	User Reminder	Entity (Database Table)
9	Content Format Controller	Controller (Format Safety Points Display)
10	Fallback Message Interface	Boundary (Alternative Flow UI)

Table 19. Use case #3 Design Object Table

3.1.3.5 Sequence Diagram #3 Review Key Safety Points

The following sequence diagram outlines the chronological flow of messages for this use case shown in the figure below.

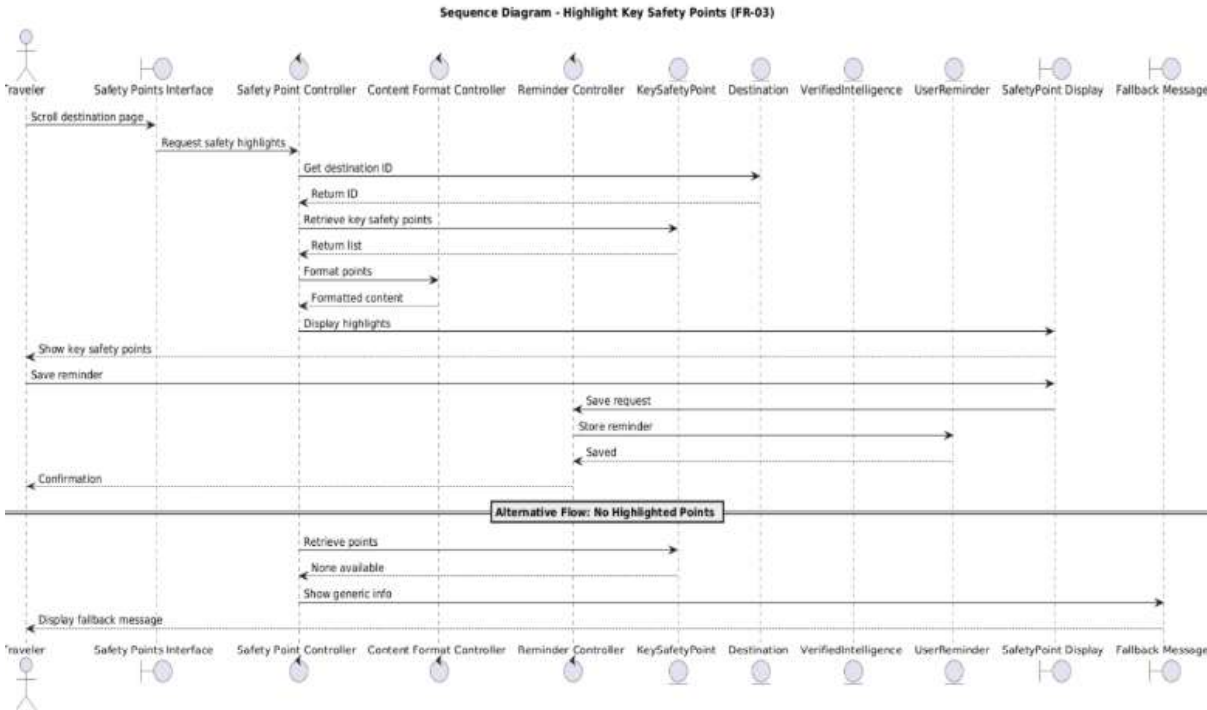


Figure 8. Use case #3 sequence diagram

3.1.4 Use Case #4 View Short Safety Tips

3.1.4.1 Use Case Scenario #4 View Short Safety Tips

Pre-Condition: The traveler has selected a destination.

Description: The system displays short safety tips, cultural notes, and awareness messages when a traveler selects a destination.

	Main Flow
1.	Actor selects a destination.
2.	Actor views the destination information page.
3.	Actor sees the sort safety tips for the selected destination.
4.	Actor reads the displayed safety tips
	Alternative Flow
1.	Actor selects a destination.

2.	Actor sees a fallback message: “Safety tips unavailable at the moment.”
3.	Actor refreshes or retries the page.
4.	Actor checks whether the safety tips appear after retrying.
5.	If the tips still do not appear, Actor retries again later.

Table 20. Use case scenario #4 flow

Post-Condition: Safety tips are displayed to the user or a fallback message is shown.

3.1.4.2 Wireframe #4 View Short Safety Tips

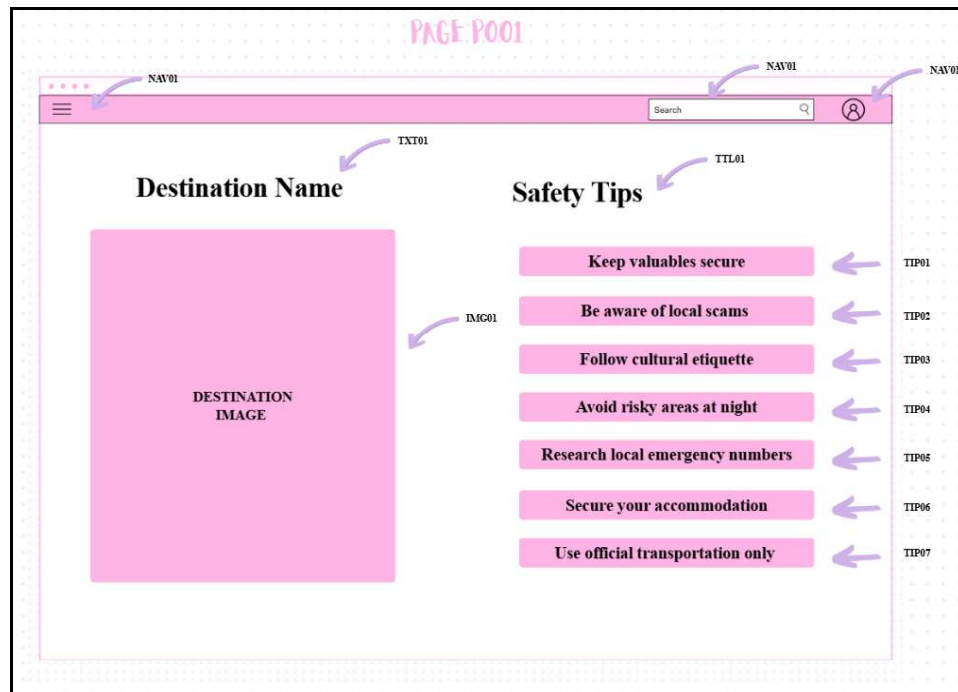


Figure 9. Use case #4 wireframe

3.1.4.3 UI Design and UI Description Object #4 View Short Safety Tips

UI Object Description Table

ID PAGE	NAME PAGE	DESCRIPTION
P001	Destination Information	Page shown after user selects a destination (displays image + safety tips)

Table 21. Use case #4 UI Object Description Table

Object ID	Type	Label	Description
NAV01	Navigation Menu	Navbar	Contains menu icon, search bar, and user icon
TXT01	Text	Destination Name	Displays the selected destination name
IMG01	Image Box	Destination Image	Shows the image for the selected destination
TTL01	Text	Safety Tips	Section header for safety tips
TIP01	Box	Keep valuables secure	Safety Tip Item
TIP02	Box	Be aware of local scams	Safety Tip Item
TIP03	Box	Follow cultural etiquette	Safety Tip Item
TIP04	Box	Avoid risky areas at night	Safety Tip Item
TIP05	Box	Research local emergency numbers	Safety Tip Item
TIP06	Box	Secure your accommodation	Safety Tip Item
TIP07	Box	Use official transportation only	Safety Tip Item

Table 22. Use case #4 interface details

3.1.4.4 Identification of New Objects and Class Type #4 View Short Safety Tips

DESIGN OBJECT TABLE

No.	New Object Name	Class Type
1.	User	Boundary
2.	Destination Information Page	Boundary
3.	Safety Tips Controller	Controller
4.	Destination	Entity

5.	Safety Tip	Entity
6.	Safety Tip List	Entity

Table 23. Use case #4 Design Object Table

3.1.4.5 Sequence Diagram #4 View Short Safety Tips

The following sequence diagram outlines the chronological flow of messages for this use case shown in the figure below.

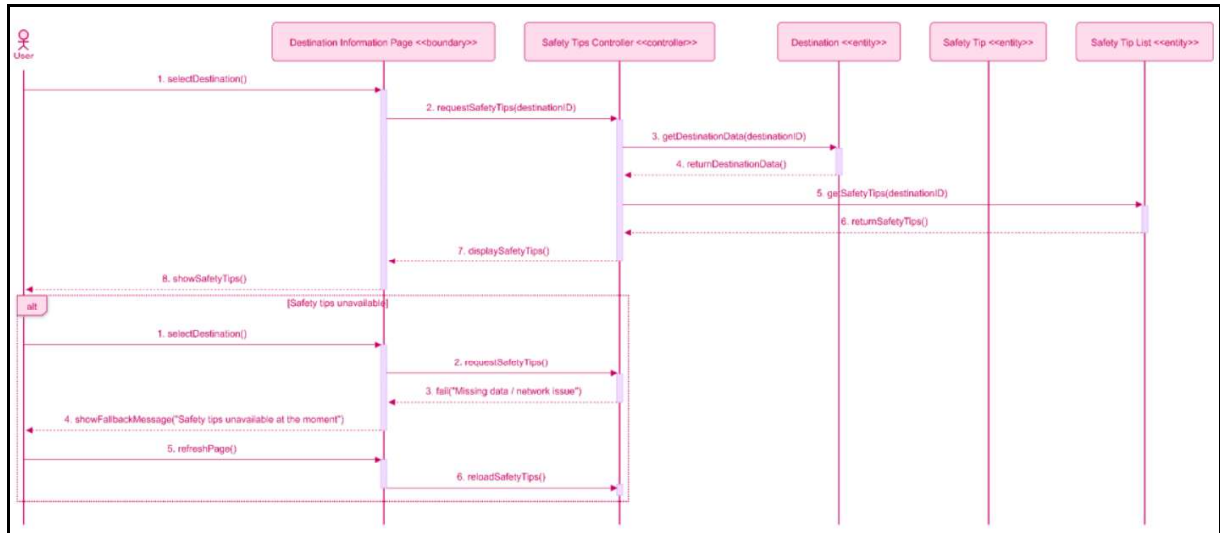


Figure 10. Use case #4 sequence diagram

3.1.5 Use Case #5 Save Safety Reminder

3.1.5.1 Use Case Scenario #5 Save Safety Reminder

Pre-Condition: The traveler is logged in and is currently viewing safety information.

Description: The system allows the traveler to save important safety reminders for future access.

	Main Flow
1.	Actor views the safety information on the destination page.
2.	Actor sees a “Save Reminder” button.
3.	Actor clicks the “Save Reminder” button.
4.	Actor sees a confirmation message: “Reminder saved successfully.”

	Alternative Flow
1.	Actor attempts to save a safety reminder.
2.	Actor sees an error message: “Failed to save reminder. Please try again.”
3.	Actor attempts to save again.
4.	Actor checks whether the reminder is saved after retrying.

Table 24. Use case scenario #5 flow

Post-Condition: The safety reminder is saved in the user account for later use.

3.1.5.2 Wireframe #5 Save Safety Reminder

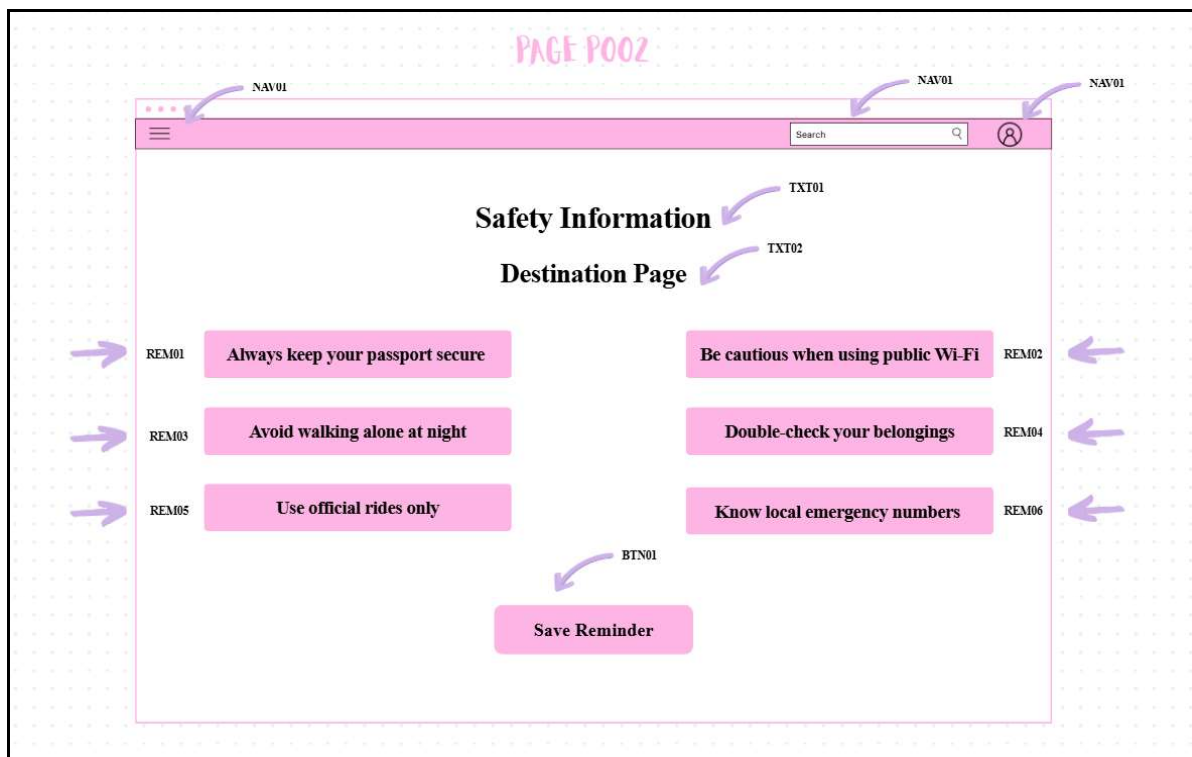


Figure 11. Use case #5 wireframe

3.1.5.3 UI Design and UI Description Object #5 Save Safety Reminder

UI Object Description Table

ID PAGE	NAME PAGE	DESCRIPTION
P001	Safety Information	Page where the user views safety information and can save safety reminders

Table 25. Use case #5 UI Object Description Table

P005 : Page Safety Information

Object ID	Type	Label	Description
NAV01	Navigation Menu	Navbar	Contains menu icon, search bar, and user icon
TXT01	Text	Safety Information	Main header of the page
TXT02	Text	Destination Page	Subheader indicating the destination context
REM01	Box	Always keep your passport secure	Safety Reminder Item
REM02	Box	Be cautious when using public Wi-Fi	Safety Reminder Item
REM03	Box	Avoid walking alone at night	Safety Reminder Item
REM04	Box	Double-check your belongings	Safety Reminder Item
REM05	Box	Use official rides only	Safety Reminder Item
REM06	Box	Know local emergency numbers	Safety Reminder Item
BTN01	Button	Save Reminder	Allows user to save the selected safety reminder

Table 26. Use case #5 interface details

3.1.5.4 Identification of New Objects and Class Type #5 Save Safety Reminder

DESIGN OBJECT TABLE

No.	New Object Name	Class Type
1.	User	Boundary
2.	Safety Information Page	Boundary
3.	Safety Reminder Controller	Controller
4.	Safety Reminder	Entity
5.	Saved Reminder	Entity
6.	User Account	Entity

Table 27. Use case #5 Design Object Table

3.1.5.5 Sequence Diagram #5 Save Safety Reminder

The following sequence diagram outlines the chronological flow of messages for this use case shown in the figure below.

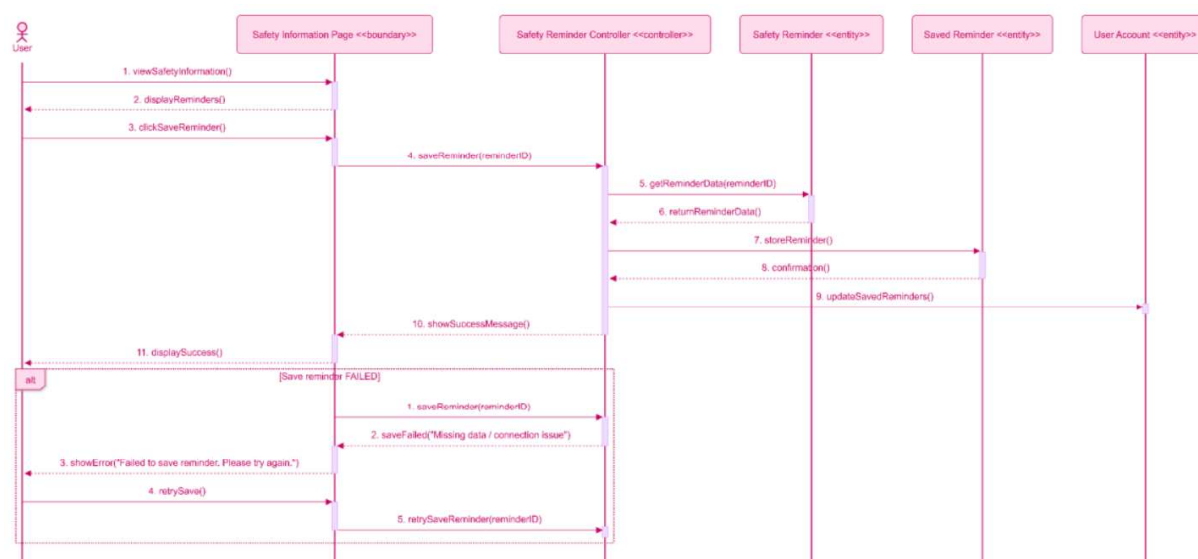


Figure 12. Use case #5 sequence diagram

3.1.6 Use Case #6 View Short Safety Quote

3.1.6.1 Use Case Scenario #6 View Short Safety Quote

Pre-Condition: The user has launched a web browser and has an active internet connection.

Description: Upon entering the website, the user is presented with a brief, awareness-promoting safety quote integrated into the homepage layout.

	Main Flow
1.	The user enters the website's main URL into the browser address bar.
2.	The user waits for the page to load and observes the main homepage content.
3.	The user sees a dedicated section displaying a specific safety quote.
	Alternative Flow
1.	The user enters the website's main URL into the browser address bar.
2.	The user views the main content of the homepage.
3.	The user observes the homepage layout without the safety quote section, allowing them to continue browsing without interruption.

Table 28. Use case scenario #6 flow

Post-Condition: The user has successfully loaded the homepage and has either read the safety quote or viewed the page content without obstruction.

3.1.6.2 Wireframe #6 View Short Safety Quote

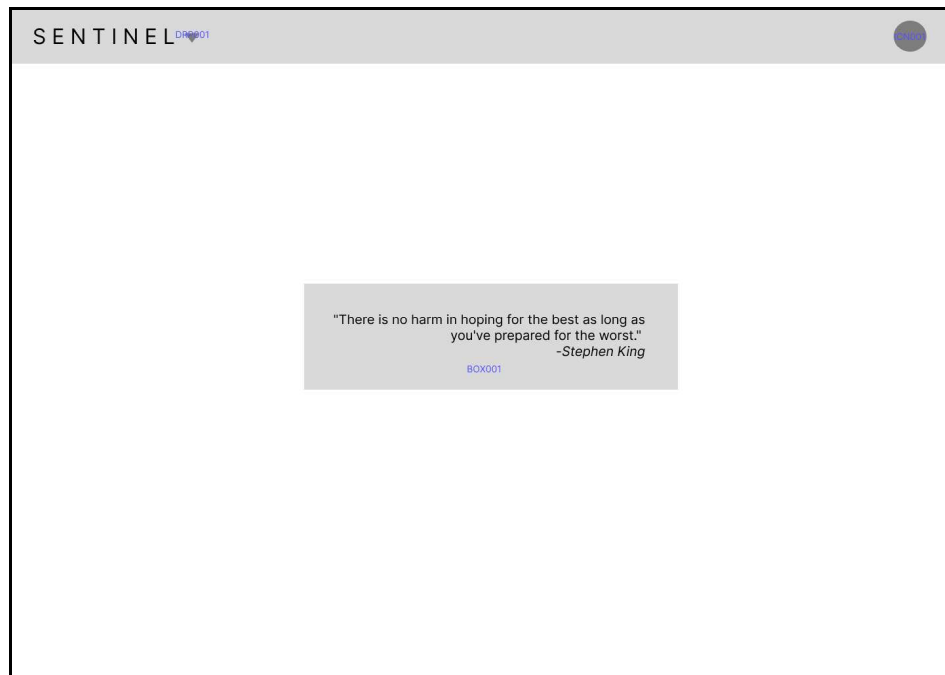


Figure 13. Use case #6 wireframe

3.1.6.3 UI Design and UI Description Object #6 View Short Safety Quote

UI Object Description Table

ID Page	Page Name	Description
P001	Home page	The landing page that displays the random safety quote.

Table 29. Use case #6 UI Object Description Table

Home Page (Display Short Safety Quotes)

<i>ID_Object</i>	<i>Type</i>	<i>Label</i>	<i>Description</i>
DRP001	Dropdown Menu	SENTINEL	The main header. When clicked, it drops down to show navigation links (e.g., Home, Map, About).
ICN001	Icon	(Circle icon)	User profile. When clicked, shows user profile information.
BOX001	Container	(Quote text)	The central box contains short safety quotes displayed to the user.

Table 30. Use case #6 interface details

3.1.6.4 Identification of New Objects and Class Type #6 View Short Safety Quote

DESIGN OBJECT TABLE

No.	New Object Name	Class Type
1.	Home Page (P001)	Boundary
2.	Safety Quote Controller	Controller
3.	Safety Quote Data	Entity

Table 31. Use case #6 Design Object Table

3.1.6.5 Sequence Diagram #6 View Short Safety Quote

The following sequence diagram outlines the chronological flow of messages for this use case shown in the figure below.

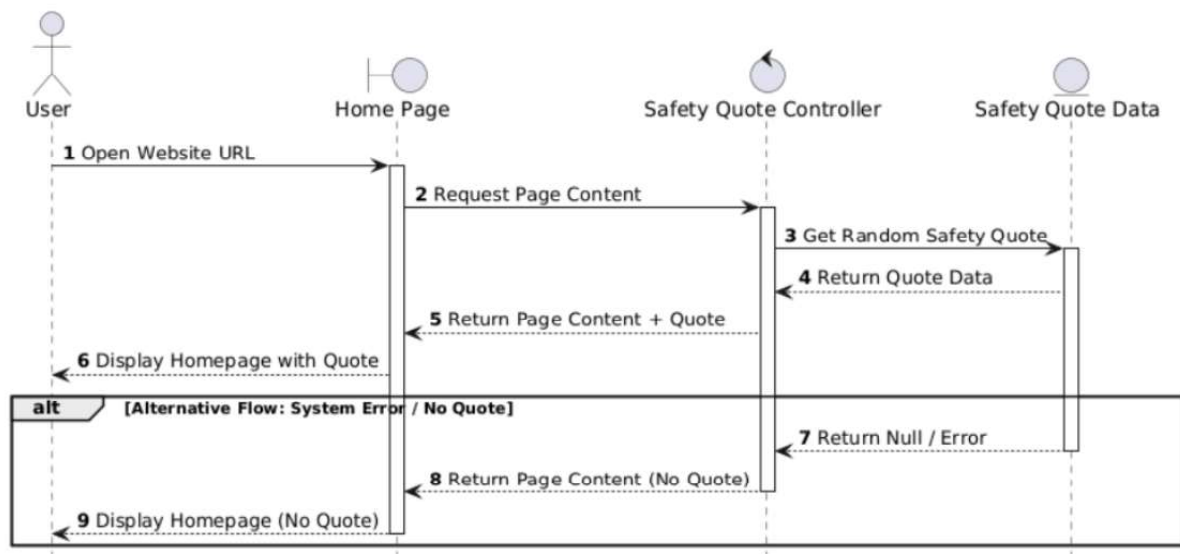


Figure 14. Use case #6 sequence diagram

3.1.7 Use Case #7 Explore Interactive Risk Map

3.1.7.1 Use Case Scenario #7 Explore Interactive Risk Map

Pre-Condition: The user has access to the internet and is currently viewing the website section designated for the risk map.

Description: The user explores a visual map interface where geopolitical risk levels are indicated by specific color-coded zones.

	Main Flow
1.	The user clicks the navigation link to access the interactive risk map.
2.	The user observes the map loading, with geopolitical areas clearly shaded in different colors to represent risk levels.
3.	The user interacts with the risk map interface, such as zooming in/out or panning across regions, and clicks or hovers to a specific colored region.
4.	Upon interaction, the user sees immediate feedback, such as the map zooming or a pop-up window appearing with text details about that specific region's information.
	Alternative Flow (Map Load Failures)
1.	The user attempts to navigate to the risk map.

2.	Instead of the risk map interface, the user sees a message stating: <i>"Map could not be loaded."</i>
3.	The user is unable to perform any map-based actions.
	Alternative Flow (Risk Data Failure)
1.	The user navigates to the risk map.
2.	The user sees the standard geographical map load successfully, but notices the color-coded risk overlays are missing.
3.	The user sees a message stating: <i>"Risk data is currently unavailable."</i>
4.	The user can zoom or pan the base map, but clicking regions does not yield risk-specific details.

Table 32. Use case scenario #7 flow

Post-Condition: The user has successfully visualized the risk distribution on the map and accessed specific details for their region of interest, or has been informed when the data is unavailable.

3.1.7.2 Wireframe #7 Explore Interactive Risk Map

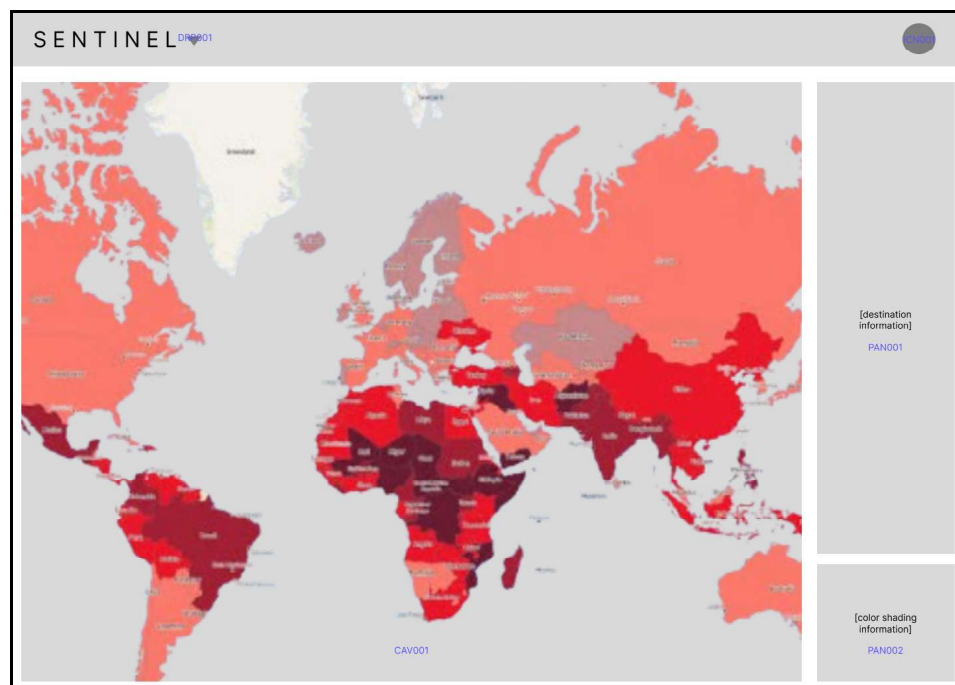


Figure 15. Use case #7 wireframe

3.1.7.3 UI Design and UI Description Object #7 Explore Interactive Risk Map

UI Object Description Table

<i>ID Page</i>	<i>Page Name</i>	<i>Description</i>
P002	Risk map page	The interactive page showing the global map and risk details.

Table 33. Use case #7 UI Object Description Table

P002: Risk Map Page

<i>ID_Object</i>	<i>Type</i>	<i>Label</i>	<i>Description</i>
DRP001	Dropdown Menu	SENTINEL	The main header. When clicked, it drops down to show navigation links (e.g., Home, Map, About).
ICN001	Icon	(Circle icon)	User profile. When clicked, shows user profile information.
CAV001	Interactive Image	N/A	The main central area shows the world map with color-coded shading, indicating risk levels.
PAN001	Content Box	[destination information]	Right sidebar section to display text details about a selected country.
PAN002	Content Box	[color shading information]	Bottom-right sidebar section explaining what each color means.

Table 34. Use case #7 interface details

3.1.7.4 Identification of New Objects and Class Type #7 Explore Interactive Risk Map

DESIGN OBJECT TABLE

No.	New Object Name	Class Type
1.	Risk Map Page (P002)	Boundary
2.	Map Controller	Controller

3.	Geopolitical Risk Data	Entity
----	------------------------	--------

Table 35. Use case #7 Design Object Table

3.1.7.5 Sequence Diagram #7 <Explore Interactive Risk Map>

The following sequence diagram outlines the chronological flow of messages for this use case shown in the figure below.

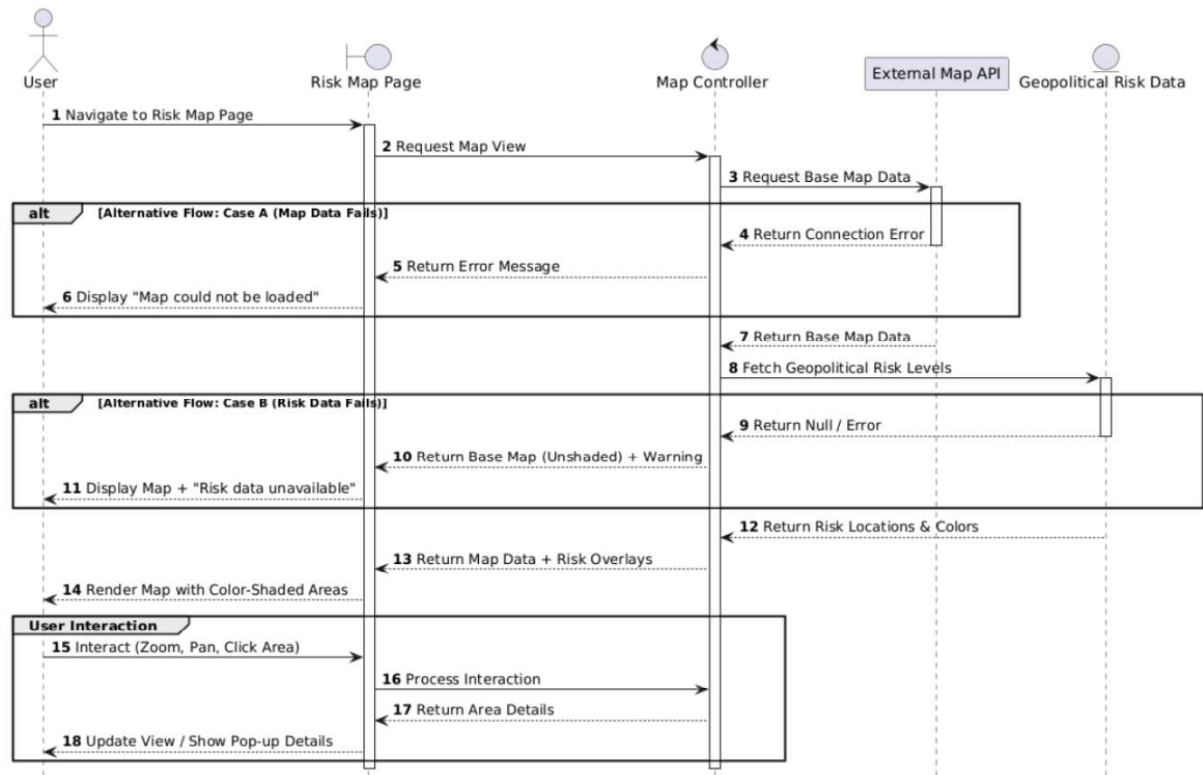


Figure 15. Use case #7 Sequence diagram

3.1.8 Use Case #8 Register New Account

3.1.8.1 Use Case Scenario #8 Register New Account

Pre-Condition: The user is currently logged out and has navigated to the registration URL.

Description: A guest user creates a new account by selecting a specific role (Traveler or Travel Agency Partner) and providing the necessary credentials.

	Main Flow
1.	The user observes the registration interface, specifically noting the option to select an account type.
2.	The user toggles or clicks to select their intended role (Traveler or Travel Agency Partner).

3.	The user fills in all required registration fields, such as name, email, password, company info if a Travel Agency Partner.
4.	The user clicks the "Register" button.
5.	The user sees a confirmation message, <i>"Registration successful"</i> , and is automatically redirected to the specific Dashboard relevant to their selected role (Traveler Dashboard or Partner Dashboard).
	Alternative Flow
1.	The user observes the registration interface, specifically noting the option to select an account type.
2.	The user toggles or clicks to select their intended role (Traveler or Travel Agency Partner).
3.	The user fills in all required registration fields, such as name, email, password, company info if a Travel Agency Partner.
4.	The user clicks the "Register" button.
5.	Instead of being redirected, the user sees the registration form reload (or stay in place) with specific fields highlighted in red (e.g., the Email field).
6.	The user reads the explanatory text next to the errors, such as <i>"Email already in use"</i> or <i>"Password must contain at least 8 characters."</i>
7.	The user changes the incorrect data and clicks the submit button again to resume the process.

Table 36. Use case scenario #8 flow

Post-Condition: The user has successfully gained access to the system as a logged-in member and is viewing their role-specific dashboard.

3.1.8.2 Wireframe #8 Register New Account

Figure 16. Use case #8 wireframe

3.1.8.3 UI Design and UI Description Object #8 Register New Account

UI Object Description Table

<i>ID Page</i>	<i>Page Name</i>	<i>Description</i>
P003	Registration page	The form page for creating a new user account.

Table 37. Use case #8 UI Object Description Table

P003: Registration Page

<i>ID_Object</i>	<i>Type</i>	<i>Label</i>	<i>Description</i>
DRP001	Dropdown Menu	SENTINEL	The main header. When clicked, it drops down to show navigation links (e.g., Home, Map, About).
ICN001	Icon		) User profile. When clicked, shows user profile information.

LBL001	Label	Register to Sentinel	The page title.
DRP002	Dropdown List	Choose Role	Allows users to select between roles.
TXT001	Text Field	Enter Name	Input field for the user's full name.
TXT002	Text Field	Enter Email Address	Input field for the user's email address.
TXT003	Text Field	Create Username	Input field for the unique username.
TXT004	Text Field	Create Password	Input field for the password (masked characters).
TXT005	Text Field	Confirm Password	Input field to re-enter password for validation. Must match the password above.
TXT006	Text Field	Other info	Input field, as additional information field (e.g. Company for Travel Agency Partner roles)
BTN001	Button	Register	Submits the form data. Only active if both passwords match.
LNK001	Hyperlink	Login	Redirects to the login page.

Table 38. Use case #8 interface details

3.1.8.4 Identification of New Objects and Class Type #8 Register New Account

DESIGN OBJECT TABLE

No.	New Object Name	Class Type
1.	Registration Page (P003)	Boundary
2.	Registration Controller	Controller
3.	User Account Data	Entity

Table 39. Use case #8 Design Object Table

3.1.8.5 Sequence Diagram #8 Register New Account

The following sequence diagram outlines the chronological flow of messages for this use case shown in the figure below.

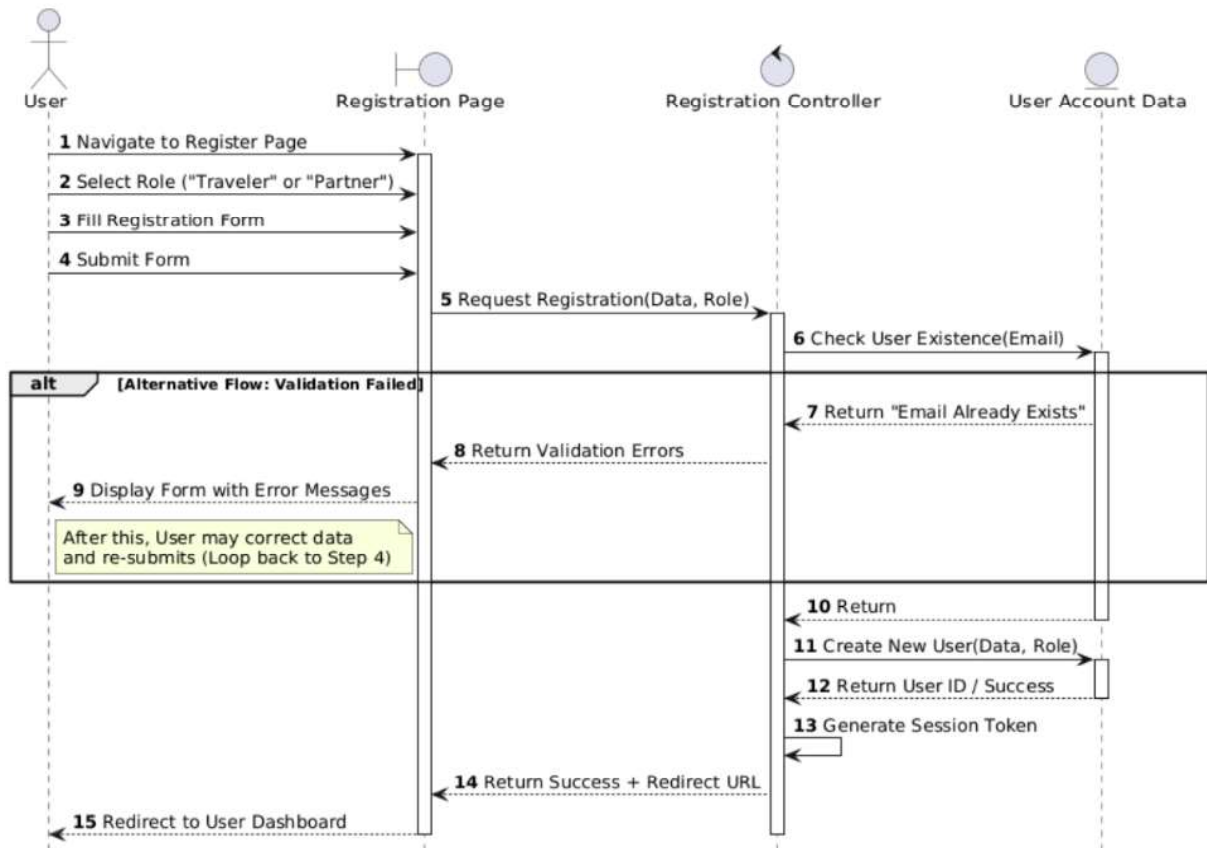


Figure 17. Use case #8 sequence diagram

3.1.9 Use Case #9 Manage User Profile

3.1.9.1 Use Case Scenario #9 Manage User Profile

Pre-Condition: The administrator should be logged into system with administrative privileges and valid credentials

Description: This use case allows the system administrator to create, update, and delete user accounts to control access, manage credentials, and ensure accurate user information.

	Main Flow
1.	The Actor (Administrator) selects the “Manage User Accounts” option.
2.	The Actor views the list of existing user accounts.
3.	The Actor chooses to create, edit, or delete a user account.

4.	The Actor accesses the appropriate interface to enter details or confirm the deletion.
	Alternative Flow
1.	The Actor enters invalid or incomplete details.
2.	The Actor reviews the error message prompting for corrections.
3.	The Actor re-enters the information with valid data.
4.	The Actor completes the operation successfully.

Table 40. Use case scenario #9 flow

Post-Condition: User account is successfully created, modified, deleted and system's database is updated accordingly

3.1.9.2 Wireframe #9 Manage User Profile

The wireframe shows the 'Manage User Accounts' interface. At the top, there's a header with 'SENTINEL — ADMIN DASHBOARD' and navigation links for 'Dashboard', 'Reports', and 'Manage User Accounts'. The main heading is 'Manage User Accounts'. Below this is a search section with a text input labeled 'Search User' and a 'Search' button. A table displays a list of users with columns for Username, Email, Role, Status, and a 'Delete' column. The table contains three rows: 'user1' (Admin, Active) with an 'Edit' button, 'user2' (User, Active) with an 'Edit' button, and 'user3' (User, Active) with an 'Edit' button. Below the table is a 'Create New User' button.

Figure 18. Use case #9 wireframe A

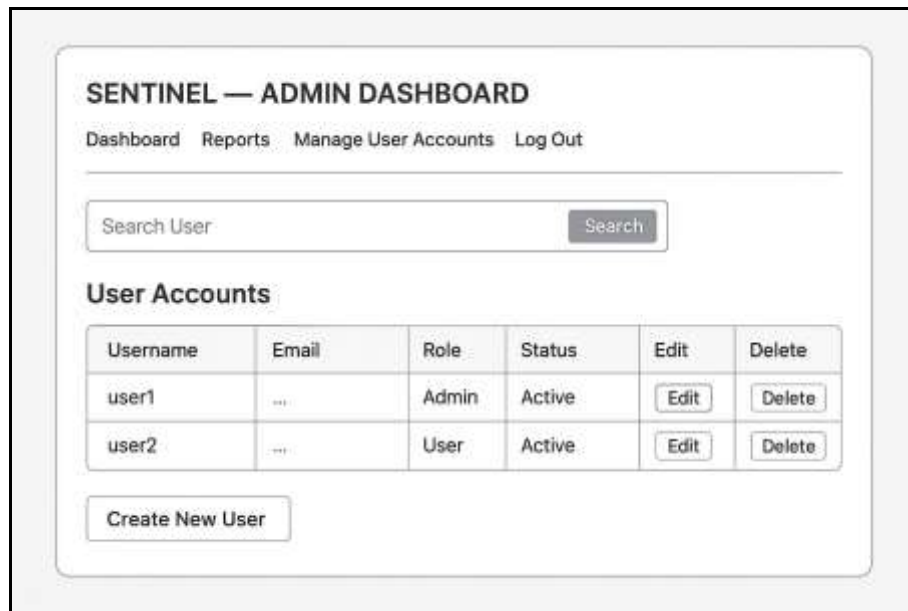


Figure 19. Use case #9 wireframe B

3.1.9.3 UI Design and UI Description Object #9 Manage User Profile

UI Object Description Table

ID	Page	Page Name	Description
P001	Manage User Accounts	Admin Page	Admin can create, edit, delete user accounts

Table 41. Use case #9 UI Object Description Table

P001 Manage User Profile

Object ID	Type	Label	Description
TXT002	Text Field	Search User	Allows admin to search users
BTN002	Button	Search	Filters user list
TBL002	Table	User Accounts	Displays user info and actions
BTN003	Button	Create User	Opens create user form

BTN00 4	Button	Edit	Opens edit user form
BTN00 5	Button	Delete	Deletes user after confirmation
MENU0 5	Menu	Dashboard	Returns to admin dashboard
MENU0 6	Menu	Reports	Opens reports page
MENU0 7	Menu	Manage User Accounts	Current page
MENU0 8	Menu	Log Out	Logs out admin

Table 42. Use case #9 interface details

3.1.9.4 Identification of New Objects and Class Type #9 <Manage User Profile>

DESIGN OBJECT TABLE

No.	New Object Name	Class Type
1.	Manage Users UI	Boundary
2.	User Account Controller	Controller
3.	User Database	Entity
4.	Authentication Module	Controller
5.	Results Table UI	Boundary

Table 43. Use case #9 Design Object Table

3.1.9.5 Sequence Diagram #9 Manage User Profile

The following sequence diagram outlines the chronological flow of messages for this use case shown in the figure below.

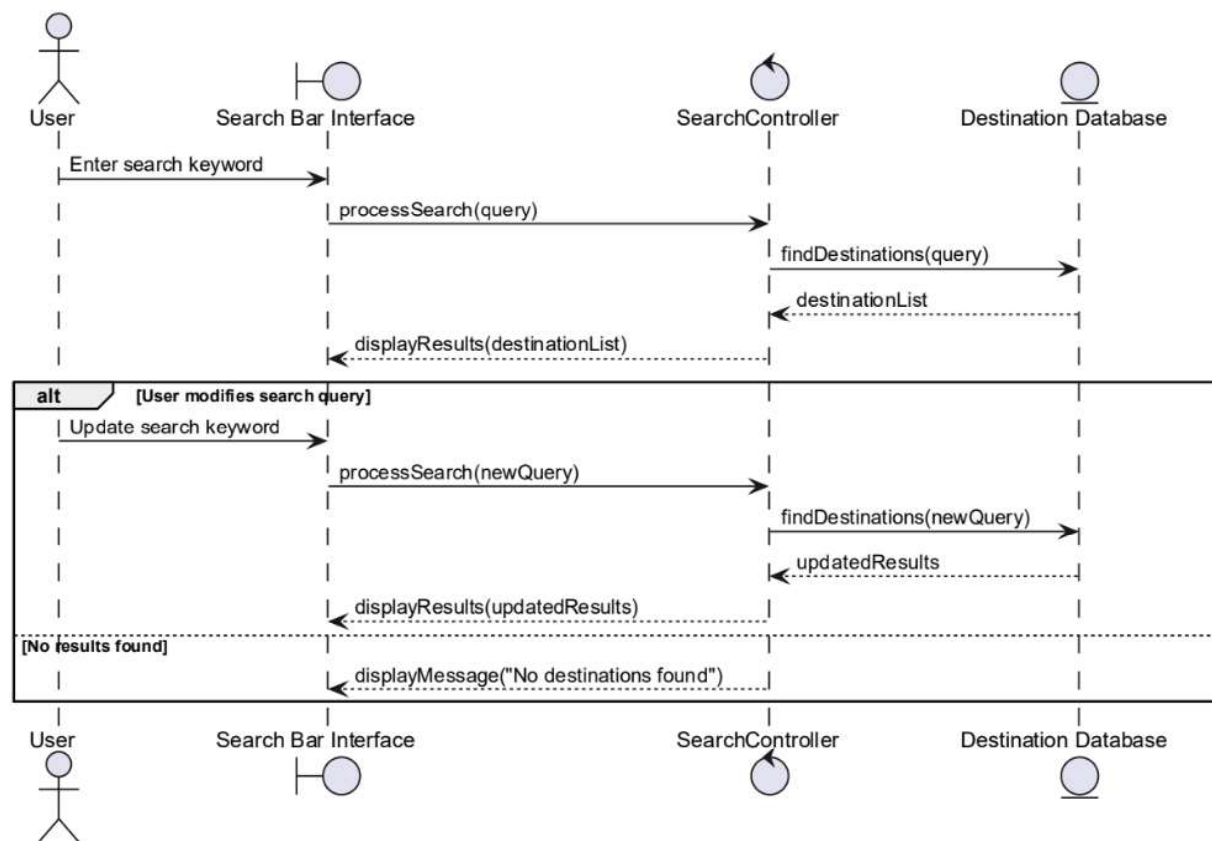


Figure 20. Use case #9 sequence diagram

3.1.10 Use Case #10 Search Bar for Searching Destinations

3.1.10.1 Use Case Scenario #10 Search Bar for Searching Destinations

Pre-Condition: The user must be logged in or on the homepage where the search bar is available.

Description: This use case enables users to search for their desired destinations using a search bar that provides relevant and quick results.

	Main Flow
1.	The Actor (User) enters the desired destination in the search bar.
2.	The Actor initiates the search by clicking the Search button or pressing Enter.
3.	The Actor views the list of destinations retrieved based on the search query.

	Alternative Flow
1.	The Actor enters an invalid or non-existing destination.
2.	The Actor observes a “No Results Found” message.
3.	The Actor modifies the search query.
4.	The Actor views updated results if available.

Table 44. Use case scenario #10 flow

Post-Condition: The system displays destination results that match the user's search query.

3.1.10.2 Wireframe #10 Search for Destination

SENTINEL TRAVEL SAFETY ADVISOR

Home Safety Alerts Destinations Login Sign Up

Enter Destination Name Search

Search Results

Destination Name	Risk Level	Safety Details	View
Paris	Low	Safe to travel	View
Tokyo	Medium	Caution advised	View

Figure 21. Use case #10 wireframe

3.1.10.3 UI Design and UI Description Object #10 Search for Destination

UI Object Description table

ID	Page	Page Name	Description
P002	Search Bar for Searching Destinations (Home Page)	Landing Page	Allows user to search destinations and see results

Table 45. Use case #10 UI Object Description Table

P002 Search for Destinations(Home Page)

Object ID	Type	Label	Description
TXT001	Text Field	Search Bar	Accepts user input for destination name; supports typing and autosuggestions.
BTN001	Button	Search	Initiates the search based on the entered destination; fetches results from database or API.
TBL001	Table	Search Results	Displays a list of destinations matching the search query; shows columns like Name, Risk Level, Safety Details, and View Details option.
LBL001	Label	No Results	Displays a message “No destinations found” if the search returns no results.
ICON001	Icon	Clear	Clears the text input in the search bar for new searches.
LOADER001	Loader	Loading Indicator	Shows loading animation while search results are being fetched.

Table 46. Use case #10 interface details

3.1.10.4 Identification of New Objects and Class Type#10 <Search for Destinations>

DESIGN OBJECT TABLE

No.	New Object Name	Class Type
1.	Home Page UI	Boundary
2.	Destination Search Bar	Boundary
3.	Search Controller	Controller

4.	Destination Database	Entity
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Table 47. Use case #10 Design Object Table

3.1.10.5 Sequence Diagram #10 Search for Destinations

The following sequence diagram outlines the chronological flow of messages for this use case shown in the figure below.

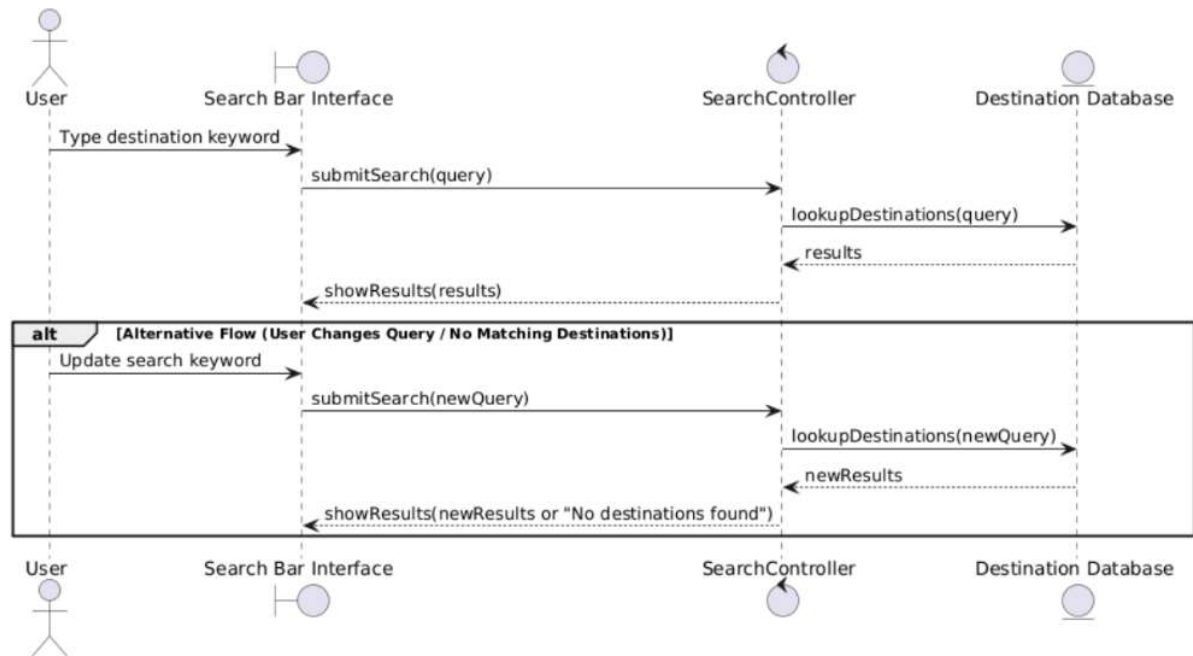


Figure 22. Use case #10 sequence diagram

3.1.11 Use Case #11 Verify Submitted Data

3.1.11.1 Use Case Scenario #11 Verify Submitted Data

Pre-Condition: The Journalist is logged into the system, The Journalist created a data and marks it, and finally The system has at least one available Risk expert on standby

Description: The system shall enable an Affiliated Journalist to submit information for verification to a Risk Expert. The system shall notify the Risk Expert of the pending verification request. The system shall allow the Risk Expert to confirm or reject the geopolitical data accuracy.

	Main Flow
1.	Journalist logs in
2.	Journalist presses the “Data verification” Button and redirected to the Data verification page
3.	Journalist uploads and submits the data and waits for verification

4.	Risk Expert got the notification
5.	Risk expert opens the notification and presses the verify data button inside the data verification notification
6.	Risk expert got redirected to the data verification page (different from the journalist's version) and must download the data to review and check before uploading back the data and either checking the "ok" or "no" buttons meaning those 2 buttons are accepted information or rejected information
	Alternative Flow
1.	Risk Expert reviews the submitted geopolitical data and finds it inaccurate, incomplete, or biased..
2.	The Risk Expert selects the option to mark the data as "Need more revision" and adds specific feedback or comments.
3.	The System updates the submission status to "Need Revision" and notifies the Affiliated Journalist.
4.	The Journalist reviews the feedback, modifies the data, and resubmits it for verification (returning to the Main Scenario).

Table 48. Use case scenario #11 flow

Post-Condition: The Journalist's notified of the outcome in which is if the data is already marked as "Verified" or "Need more revision"

3.1.11.2 Wireframe #11 Verify Submitted Data

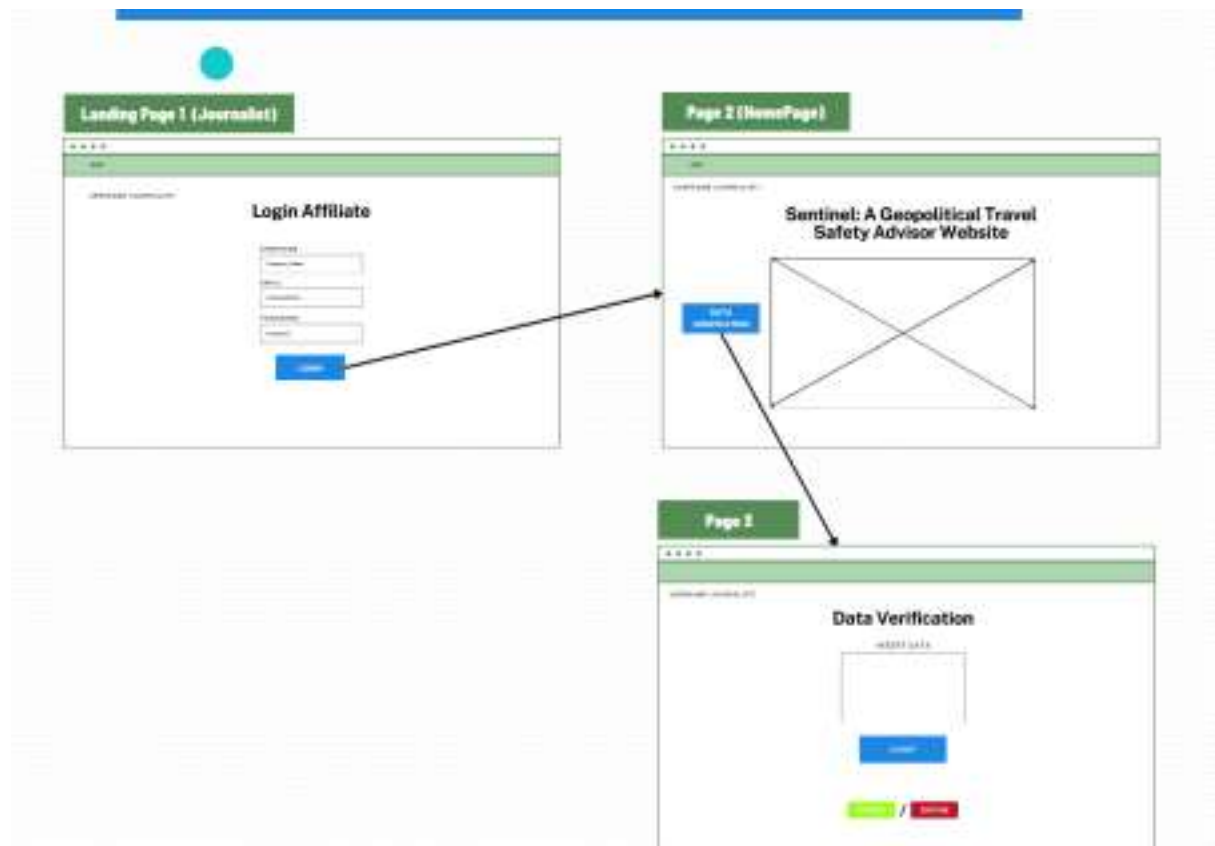


Figure 23. Use case #11 wireframe

3.1.11.3 UI Design and UI Description Object #11 Verify Submitted Data

UI Object Description Table

ID.Page	Name Page	Description
P001	Journalist Login Page	It's a login page for the journalists
P002	Homepage	It's the homepage
P003	Data verification page	It's the page where the
		journalist uploads the data and waits for the verification

P004	Risk expert homepage	The home page for risk experts
P005	Risk expert's notification	The viewed risk expert's notification
P006	Data Verification page	The part where the risk expert's task to check

Table 49. Use case #11 UI Object Description Table

Homepage (P002)

Object_ID	Type	Label	Description
menu	Title	menu	The title of the website
Data verification	button	Data verification	When the button is pressed the user is redirected to another page
The big X	A group of UI	The big X	A group of the UI's that i dont have the energy to make

Table 50. Use case #11 interface details

Journalist Login Page (P001)

Object_ID	Type	Label	Description
username	textbox	username	Type in the username
email	textbox	email	Type in the email
password	textbox	password	Type in the password
Login	button	login	When the presses button it checks with the database

			and then it logged in
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Table 51. Use case #11 interface details

Data verification page (P003)

Object_ID	Type	Label	Description
title	menu	Title	Title of the page
insertData	insertbox	Insert data	The place for inserting data
button1	button	submit	The button shall submit the data
text	status	text	Green means accepted and red means rejected

Table 52. Use case #11 interface details

Risk expert homepage (P004)

Object_ID	Type	Label	Description
menu	Title	menu	The title of the website
Data verification	button	Data verification	When the button is pressed the user is redirected to another page
The big X	A group of UI	The big X	A group of the UI's that i dont have the energy to make
notification	button	notification	Opens the notification

Table 53. Use case #11 interface details

Risk expert's notification (P005)

menu	Title	menu	The title of the website
Data verification	button	Data verification	When the button

			is pressed the user is redirected to another page
Data verification	button	Data verification	When the button is pressed the user is redirected to another page
menu	Title	menu	The title of the website
notification	button	notification	With notification

Dropdown	textbox	dropdown	Notification with the data
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Data verification page (P006)

Object_ID	Type	Label	Description
title	menu	Title	Title of the page
insertData	insertbox	Insert data	The place for inserting data
button1	button	submit	The button shall submit the data
text	status	text	Green means accepted and red means rejected

3.1.11.4 Identification of New Objects and Class Type #11 <Verify Submitted Data>

DESIGN OBJECT TABLE

No.	New Object Name	Class Type
1	Journalist login page	Boundary
2	Homepage	Entity
3	Data verification	Controler
4	Homepage with notification	Controler
5	Data verification (Risk expert)	Controler

3.1.11.5 Sequence Diagram #11 <Verify Submitted Data>

The following sequence diagram outlines the chronological flow of messages for this use case shown in the figure below.

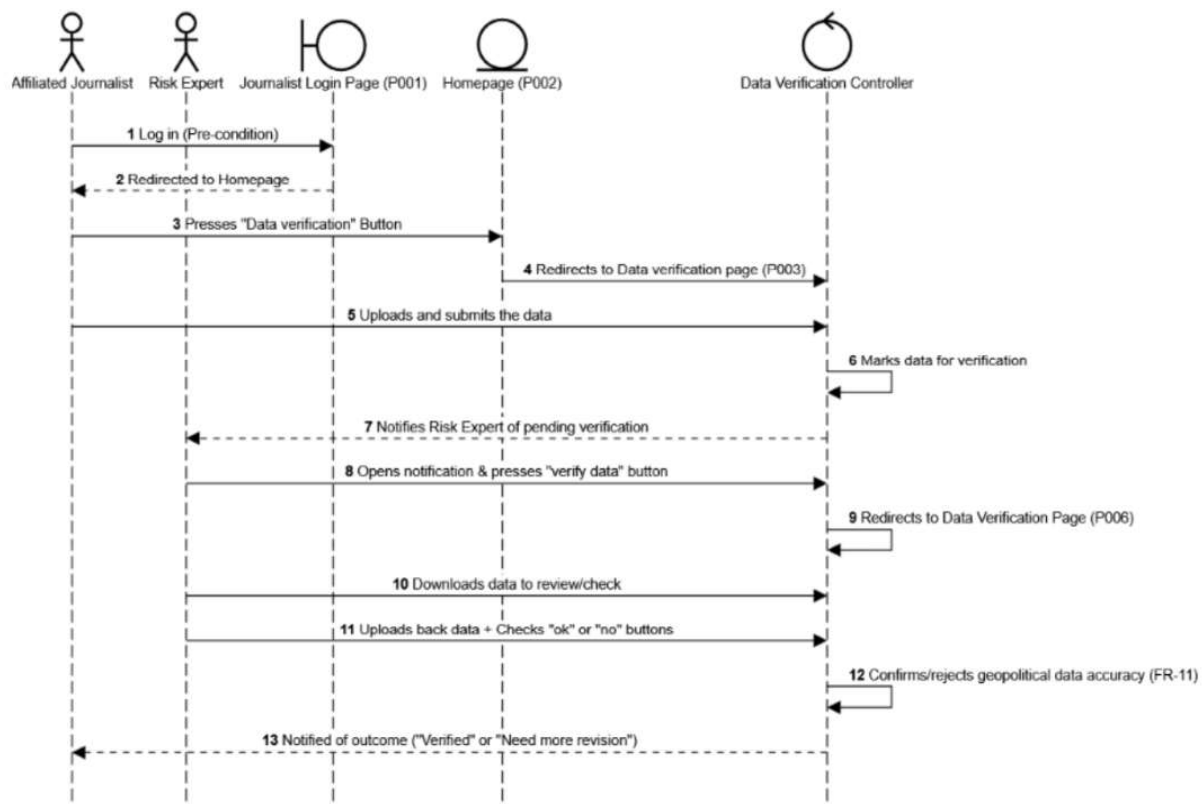


Figure 3.1.11.4.1 Sequence Diagram #11

3.1.11.6 Use Case #12 <Filter and Sort Results>

3.1.11.7 Use Case Scenario #12 <Filter and Sort Results>

Pre-Condition: The user is viewing a page that displays a list of news articles or destinations, each item in the list has a defined risk level (Low, Medium, or High).

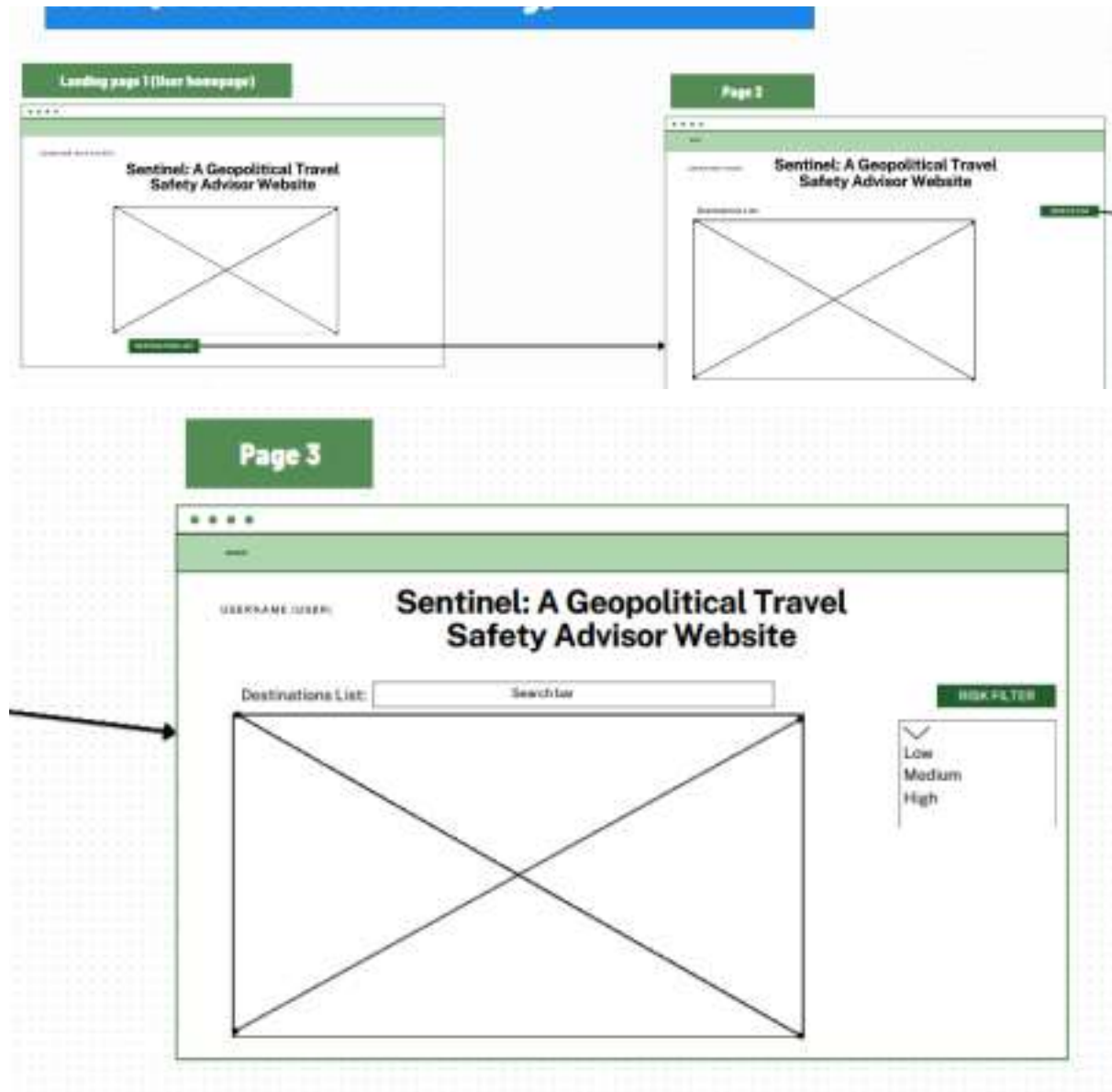
Description: The system should allow users to filter and sort destinations or news articles based on risk levels, such as low, medium, or high.

	Main Flow
1.	The user in the homepage presses the Destination List button.
2.	The user sees the destination lists
3.	The user can use the search bar to see the destination lists searched
	Alternative Flow
1.	The User applies a specific combination of filters (e.g., Risk Level: "High" + specific Region).
2.	The System executes the query and finds zero destinations or news articles matching those exact criteria.

3.	The System displays a "No items match your selected filter criteria" message.
4.	The System suggests that the user broaden their search or reset the filters to see all available data.

Post-Condition: The User is authenticated, logged in, and successfully directed to their role-specific landing page. A session is established.

3.1.11.8 Wireframe #12 <Filter and Sort Results>



3.1.11.9 UI Design and UI Description Object #12 <Filter and Sort Results>

UI Object Description Table

ID.Page	Name Page	Description
---------	-----------	-------------

P001	Homepage	It's a login page for the journalists
P002	Homepage1	It's the homepage
P003	Homepage2	Filters and search

Homepage (P001)

Object_ID	Type	Label	Description
menu	Title	menu	The title of the website
Data verification	button	Data verification	When the button is pressed the user is redirected to another page
The big X	A group of UI	The big X	A group of the UI's that i dont have the energy to make
Destination List	button	DestinationList	Redirects to the destination list page

Homepage2 (P002)

Object_ID	Type	Label	Description
menu	Title	menu	The title of the website
Data verification	button	Data verification	When the button is pressed the user is redirected to another page

The big X	A group of UI	The big X	A group of the UI's that i dont have the energy to make
Destination List	button	DestinationList	Redirects to the destination list page
RiskFilter	button	RiskFilter	Opens the filter bar

Homepage3 (P003)

Object_ID	Type	Label	Description
menu	Title	menu	The title of the website
Data verification	button	Data verification	When the button is pressed the user is redirected to another page
The big X	A group of UI	The big X	A group of the UI's that i dont have the energy to make
Destination List	button	DestinationList	Redirects to the destination list page
RiskFilter	button	RiskFilter	Opens the filter bar
Search bar	textinformation	searchbar	Search bar

3.1.11.10 Identification of New Objects and Class Type#12 <Filter and Sort Results>

DESIGN OBJECT TABLE

No.	New Object Name	Class Type
1	Homepage	Boundary
2	Homepage with button	Controler
3	Search data and filter	Entity

3.1.11.11 Sequence Diagram #12 <Data Filtering & Sorting>

The following sequence diagram outlines the chronological flow of messages for this use case shown in the figure below.

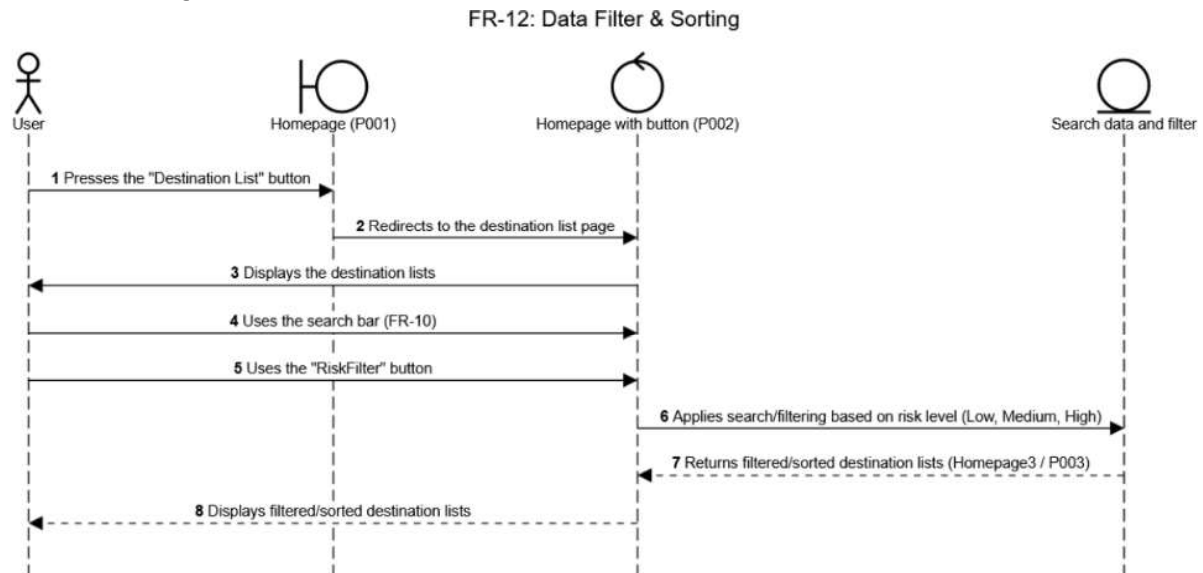


Figure 3.1.12.4.1 Sequence Diagram #12

3.1.12 Use Case #13 <Generate Travel Route>

3.1.12.1 Use Case Scenario #13 <Generate Travel Route>

Pre-Condition: The travel agency must be affiliated with the website.

Description: Travel agency partners will have access to an intelligent route-planning feature that suggests optimal routes and destinations based on geopolitical conditions and safety data.

	Main Flow
1.	The Travel agency is logged in to the authorised account.
2.	The Travel agency goes to a specialized menu for travel agencies
3.	The Travel agency will view the Travel Route menu
4.	The Travel agency will search for the location they want to find a route for
5.	The Travel Agency will view the map of the location

6.	The Travel Agency will activate “Generate route”
7.	The Travel agency will view the optimal route for the location based on geopolitical conditions and safety data
8.	The Travel agency will view information and news regarding the location
9.	The Travel Agency can also request an alternative route
10.	The Travel Agency will view an alternative route
11.	The Travel Agency can save the route and information
12.	The Travel Agency can use the route and information for their agency
	Alternative Flow
1.	The Travel Agency is logged in to the authorised account.
2.	The Travel Agency goes to a specialized menu for travel agencies
3.	The Travel Agency will view the Travel Route menu
4.	The Travel Agency will search for the location they want to find a route for
5.	The Travel Agency will view a map of the location
6.	The Travel Agency will activate “Generate route”
7.	The Travel Agency will view that currently there are no safe available route.

Post-Condition: The Travel Agency will have access to an optimal route provided by the website

3.1.12.2 Wireframe #13 <Generate Travel Route>



3.1.12.3 UI Design and UI Description Object #13 <Generate Travel Route>

UI Description Table

ID	Page	Page Name	Description
P001	Home Page	Landing Page	The initial screen displaying the main menu and navigation to the Travel Route Assistance feature
P002	Login Page	Login Form	Page where users enter email & password to access their account
P003	Travel Route Assistance Page	Generate Route	Main page for searching destinations, viewing recommendations, and generating a travel route
P004	Route Result Page	Route Output	Page displaying the generated travel route

P001: Home page

Object ID	Label Type	Description
BTN01	Button — Login	Redirects user to the Login Page
MENU01	Menu — Tourism News	Navigates to tourism-related news
MENU02	Menu — Travel Agency	Navigates to travel agency services

P002: Login Page

Object ID	Label Type	Description
TXT01	Textfield — Email	User email input
TXT02	Textfield — Password	User password input
BTN02	Button — Log In	Validates login and enters the system
LBL01	Label — “Log in to Your Account”	Page title for login

P003: Generate Travel Route Page

Object ID	Label Type	Description
TXT03	Textfield — Destination Search	Destination input field
BTN03	Button — Generate Route	Starts the route generation process
SEC01	Section — Popular	Displays featured destinations (Paris, Tokyo,

Object ID	Label Type	Description
TXT04	Destinations	Ubud, etc.)
BTN04	Button — Add Notes/Comments	Adds travel notes

MENU03	Menu — Recent News	Shows latest news related to the destination
--------	--------------------	--

P004: Route Result Page

Object ID	Label Type	Description
MAP01	Display — Generated Route	Visual map of the generated route
BTN05	Button — Regenerate Route	Provides alternative travel route options
SEC02	Section — Route Information	Shows route details
BTN06	Button — Save Route	Saves the generated route to user account

3.1.12.4 Identification of New Objects and Class Type #13 <Generate Travel Route>

DESIGN OBJECT TABLE

No.	New Object Name	Class Type
1	User	Actor
2	Login Page	Boundary
3	Travel Route Assistance Page	Boundary
4	Destination Input Field	Boundary
5	Generate Route Button	Boundary
6	RouteController	Controller
7	Destination Database	Entity
8	Route Calculation Service	Controller
9	Cities / Destination Data	Entity
10	Route Result Page	Boundary
11	News Fetching Service	Controller
12	User Account Data	Entity

3.1.12.5 Sequence Diagram #13 <Generate Travel Route>

The following sequence diagram outlines the chronological flow of messages for this use case shown in the figure below.

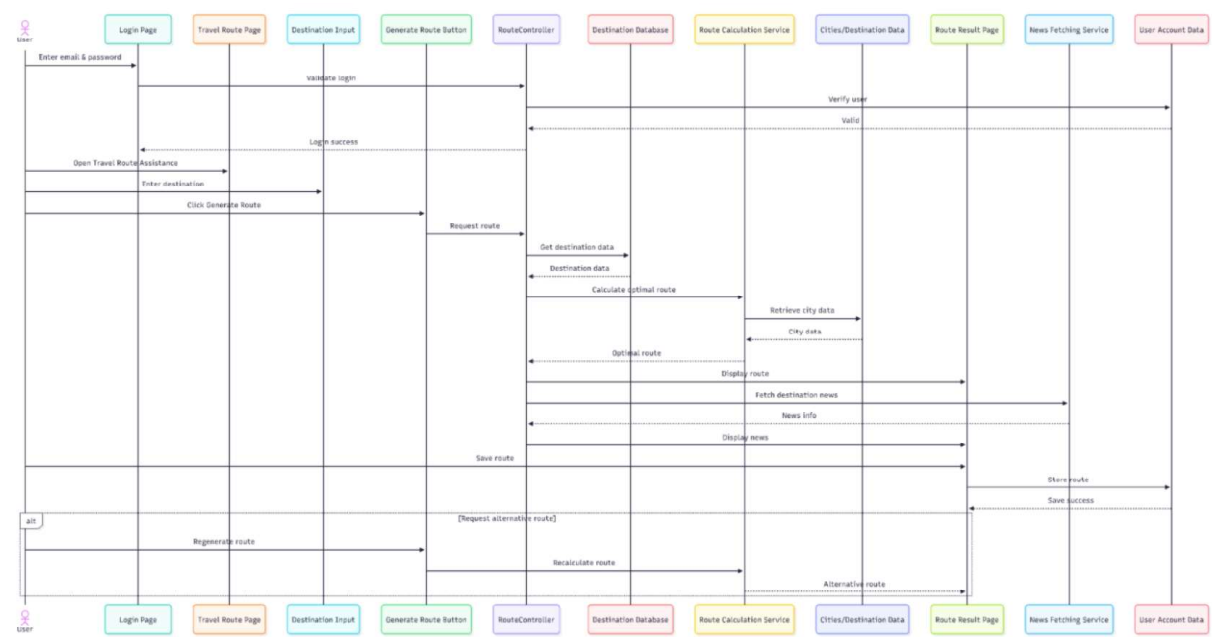


Figure 3.1.13.4.1 Sequence Diagram #13

3.1.13 Use Case #14 <Administer System Data>

3.1.13.1 Use Case Scenario #14 <Administer System Data>

Pre-Condition: The actor is connected to the internet and the actor has logged in to the authorised account.

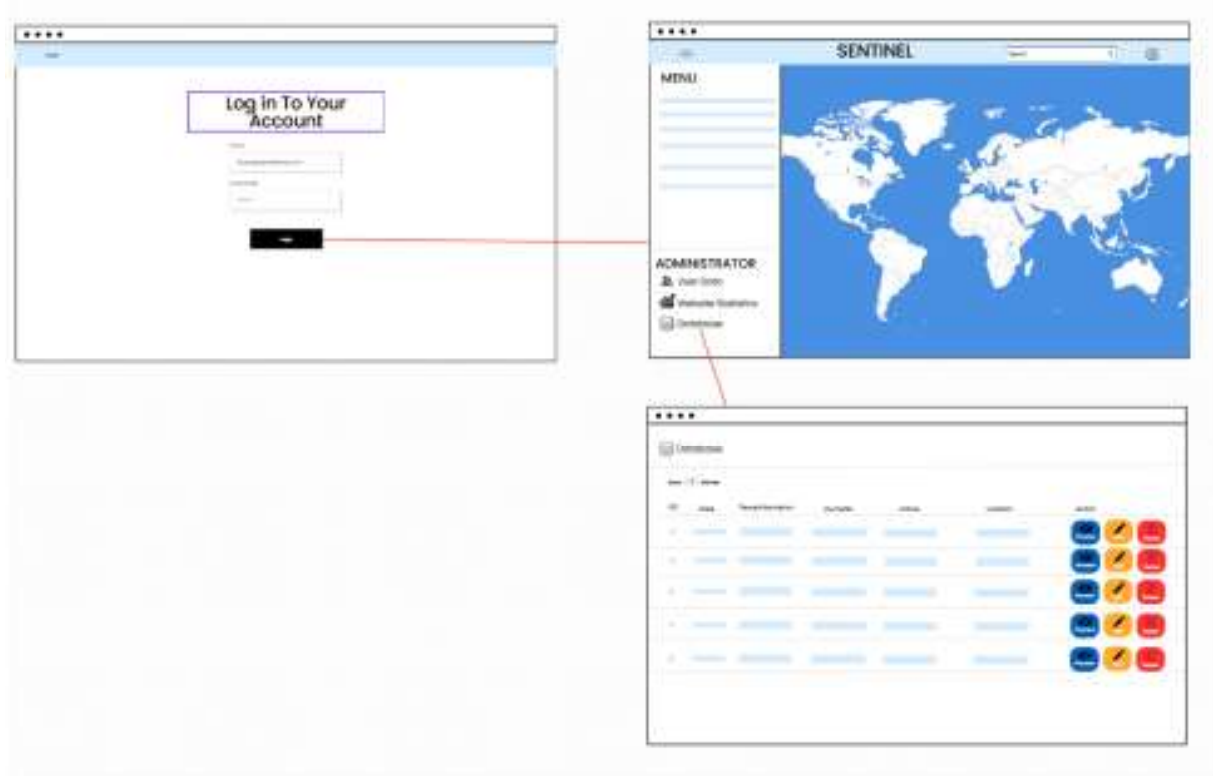
Description: Administrators have the authority to manage database content, including adding, updating, and deleting data related to users, destinations, and safety information.

	Main Flow
1.	The administrator is logged in to the authorized account
2.	The administrator accesses the database
3.	The administrator will be notified if the account has access
4.	The administrator will give access if it is a verified account
5.	The administrator will view the database
6.	The administrator will have access to manage the database
7.	The administrator can freely edit data in the database
8.	The administrator will wait for verification for the edit

9.	The administrator will view that the content of the database has been changed
	Alternative Flow
1.	The administrator is logged in to the authorized account
2.	The administrator accesses the database
3.	The administrator will be notified if the account has access
4.	The administrator doesn't get verification for the account
5.	The administrator will view a display of "User doesn't have access"

Post-Condition: The actor now has access to the database and the actor can edit data in the database.

3.1.13.2 Wireframe #14 <Administer System Data>



3.1.13.3 UI Design and UI Description Object #14 <Administer System Data>

UI Description Table

ID	Page	Page Name	Description
P005	Login Page	Admin Login	Page where administrator enters email and password

			to access the system
P006	Admin Dashboard	Main Menu	Central admin interface showing navigation to system modules Displays list of registered users and allows updating or deleting
P007	News Management Page	News/Information Table	Admin page for managing news/articles, including preview, edit, delete

P005: Login Page

Object ID	Label Type	Description
TXT01	Textfield — Email	Field for admin email input
TXT02	Textfield — Password	Field for admin password
BTN01	Button — Log In	input Authenticates admin
LBL01	Label — “Log in to Your Account”	Login page heading

P006: Admin Dashboard

Object ID	Label Type	Description
LBL02	Label — SENTINEL	System/website title
MENU01 Menu	Menu — Administrator	Admin profile/navigation
MENU02 Menu	Menu — User Data	Opens User Management page
MENU03 Menu	Menu — Website Statistics	tics Opens analytics page

MENU04 Menu	Menu — Database	Opens database-related management
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P007: News Management Page

Object ID	Label Type	Description
TAB02	Table — News/Information	Displays list of news articles
COL01	Table Column — Date	Shows publication date
COL02	Table Column — News/Information	Article title
COL03	Table Column — Journalist	Author name
COL04	Table Column — Article Location	Source or category
BTN05	Button — Preview	Opens article preview
BTN06	Button — Edit	Opens article editor
BTN07	Button — Delete	Removes article

3.1.13.4 Identification of New Objects and Class Type#14 <Administer System Data>

DESIGN OBJECT TABLE

No.	New Object Name	Class Type
1	Administrator	Actor
2	Admin Login Page	Boundary
3	Admin Dashboard	Boundary
4	User Data Page	Boundary
5	News Management Page	Boundary
6	Email Input Field	Boundary

7	Password Input Field	Boundary
8	Login Button	Boundary
9	ContentManagementControl ler	Controller
10	User Database	Entity
11	News/Articles Database	Entity
12	StatisticsController	Controller
13	Website Analytics Data	Entity
14	Preview/Edit/Delete Buttons	Boundary

3.1.13.5 Sequence Diagram #14 <Administer System Data>

The following sequence diagram outlines the chronological flow of messages for this use case shown in the figure below.

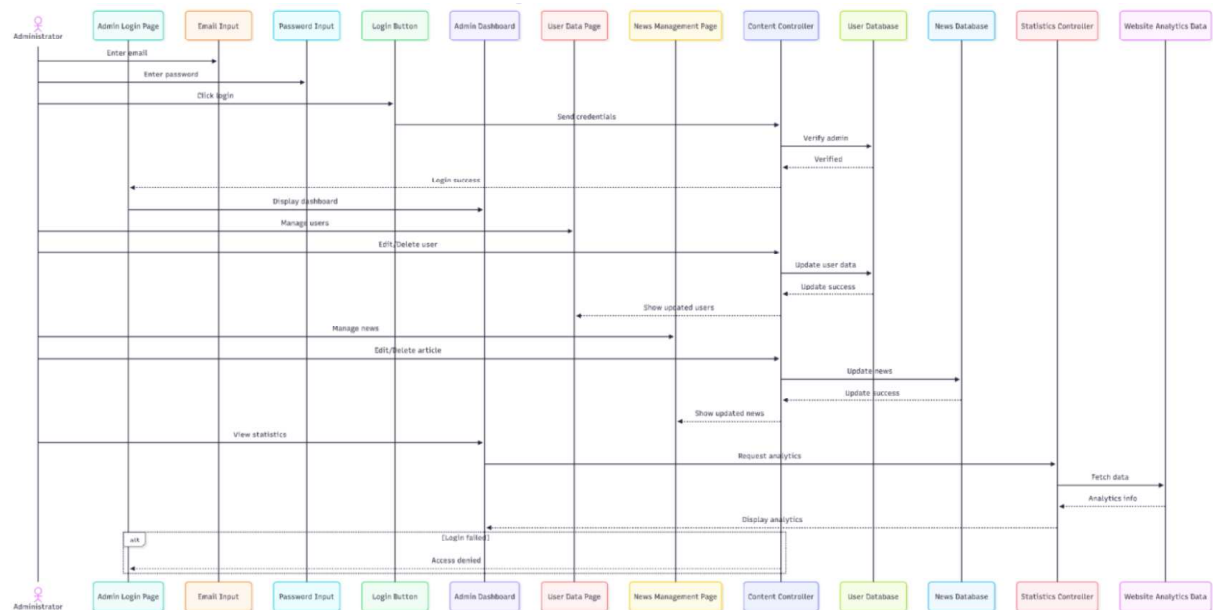


Figure 3.1.14.4.1 Sequence Diagram #14

3.1.14 Use Case #15 <Log into Application>

3.1.14.1 Use Case Scenario #15 <Log into Application>

Pre-Condition: The User has a registered account and valid login credentials, the User is viewing the Login Page, the system is connected to the user database.

Description: The system shall provide separate login and registration functionality for the Traveler role. The system shall provide separate login and registration functionality for the Travel Agency Partner role.

	Main Flow
1.	User opens the website
2.	User inputs their username, email, password
3.	After a successful log in the user is redirected to the homepage
	Alternative Flow
1.	The User enters an incorrect email address or password in the login form.
2.	The User is waiting for the output of the system.
3.	The user sees the displayed error message "Invalid email or password. Please try again."
4.	The User remained on the login page and prompted to re-enter their details or use the "Forgot Password" feature.

Post-Condition: The User is authenticated, logged in, and successfully directed to their role-specific landing page. A session is established.

3.1.14.2 Wireframe #15 <Log into Application>



3.1.14.3 UI Design and UI Description Object #15 <Log into Application>

UI Description Design Table

ID.Page	Name Page	Description
P001	Login Page	It's a login page for the user

P002	Homepage	It's the homepage
------	----------	-------------------

Login Page (P001)

Object_ID	Type	Label	Description
username	textbox	username	Type in the username
email	textbox	email	Type in the email
password	textbox	password	Type in the password
Login	button	login	When the presses button it checks with the database and then it logged in

Homepage (P002)

Object_ID	Type	Label	Description
username	textbox	username	Type in the username
email	textbox	email	Type in the email
password	textbox	password	Type in the password
Login	button	login	When the presses button it checks with the database and then it logged in

3.1.14.4 Identification of New Objects and Class Type #15 <Log into Application>

DESIGN OBJECT TABLE

No.	New Object Name	Class Type
1	login page	Boundary
2	Homepage	Entity

3.1.14.5 Sequence Diagram #15 <Log into Application>

The following sequence diagram outlines the chronological flow of messages for this use case shown in the figure below.

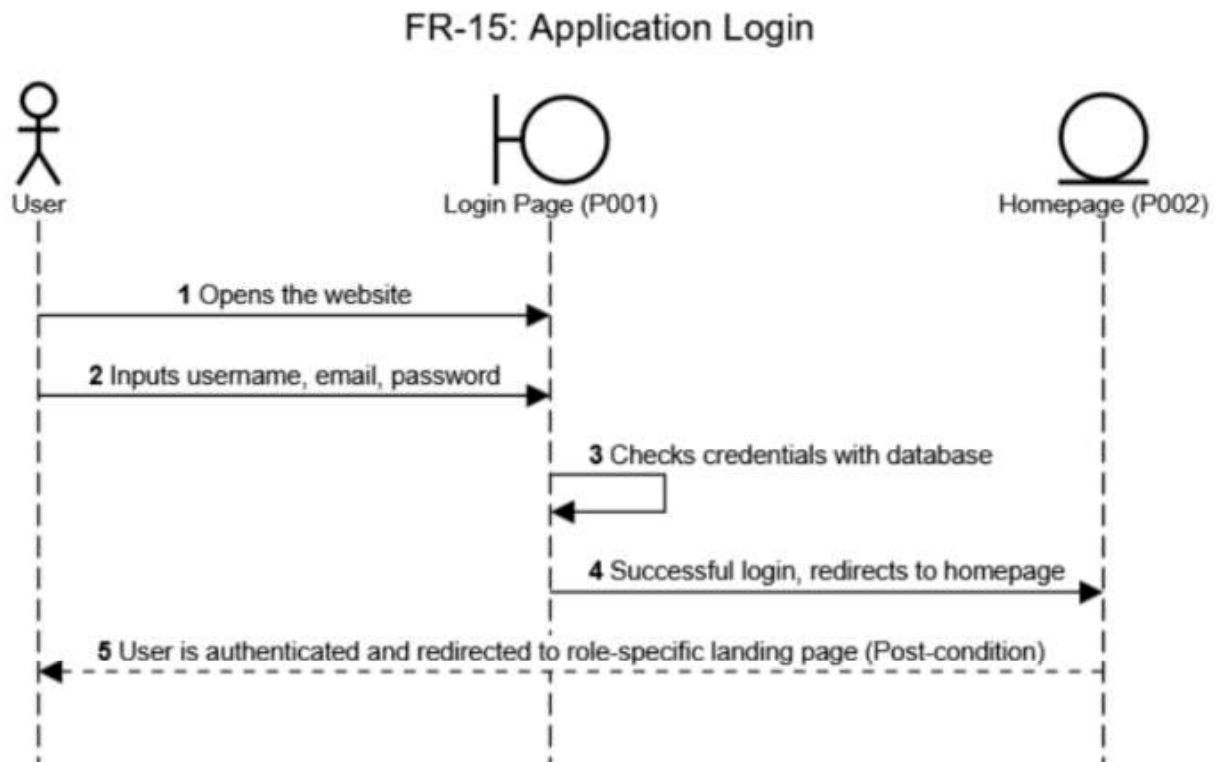


Figure 3.1.15.4.1 Sequence Diagram #15

3.2 Overall Class Diagram

The class diagram for the Sentinel website is shown in Figure 3.2.1.

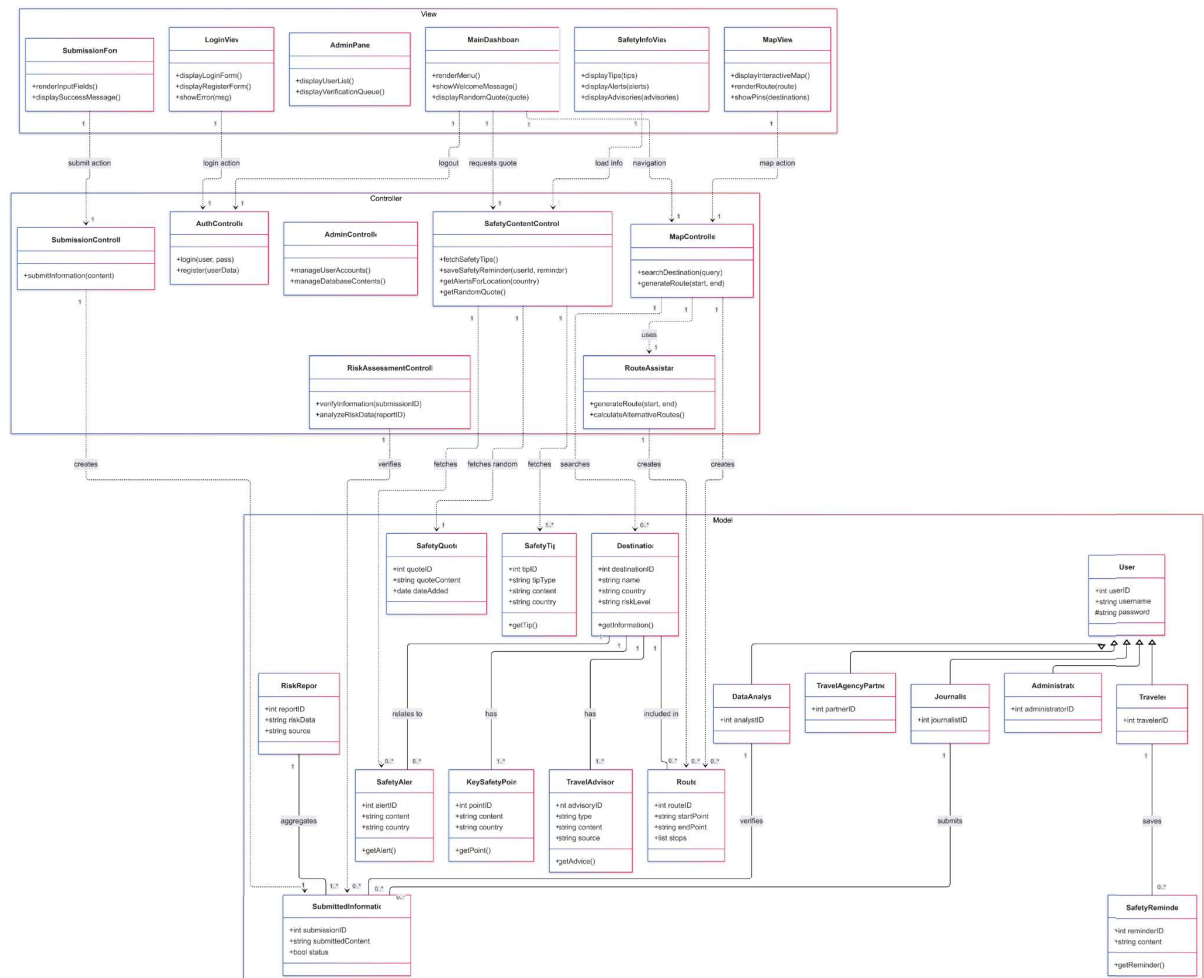


Figure 3.2.1 Class diagram

3.3 Detailed Class Diagram

Table 3.3.1 defines each class along with its respective attributes and operations.

Class ID	Design Class Name	Attributes (visibility)	Methods / Operations
Model - Class Table			
C1	User	+userID : int +username : string #password : string	
C2	Traveler	+travelerID : int	
C3	Administrator	+administratorID : int	
C4	Travel Agency Partner	+partnerID : int	
C5	Data Analyst	+analystID : int	
C6	Journalist	+journalistID : int	
C7	Destination	+destinationID : int +name : string +country : string +riskLevel : string	+getInformation() : string
C8	Safety Tip	+tipID : int +tipType : string +content : string +country : string	+getTip() : SafetyTip
C9	Safety Reminder	+reminderID : int +content : string	+getReminder() : SafetyReminder
C10	Safety Alert	+alertID : int +content : string +country : string	+getAlert() : SafetyAlert
C11	Route	+routeID : int +startPoint : string +endPoint : string +stops : list	
C12	Submitted Information	+submissionID : int +submittedContent : string +status : bool	
C13	Risk Reports	+reportID : int +riskData : string +source : string	
C14	Travel Advisory	+advisoryID : int	+getAdvice() : TravelAdvisory

		+type : string +content : string +source : string	
C15	Key Safety Point	+pointID : int +content : string +country : string	+getPoint() : KeySafetyPoint
C16	Safety Quote	+quoteID : int +quoteContent : string +dateAdded : date	
View - Class Table			
C17	Login View		+displayLoginForm() : void +displayRegisterForm() : void +showError(msg) : void
C18	Main Dashboard		+renderMenu() : void +showWelcomeMessage() : void +displayRandomQuote(quote) : void
C19	Map View		+displayInteractiveMap() : void +renderRoute(route) : void +showPins(destinations) : void
C20	Safety Info View		+displayTips(tips) : void +displayAlerts(alerts) : void +displayAdvisories(advisories) : void
C21	Admin Panel		+displayUserList() : void +displayVerificationQueue() : void
C22	Submission Form		+renderInputFields() : void +displaySuccessMessage() : void
Controller - Class Table			
C23	Auth Controller		+login(user, pass) : bool +register(userData) : void
C24	Map Controller		+searchDestination(query) : list<Destination> +generateRoute(start, end) : Route
C25	Safety Content Controller		+fetchSafetyTips() : list<SafetyTip> +saveSafetyReminder(userId, reminder) : void

			+getAlertsForLocation(country)): list<SafetyAlert> +getRandomQuote() : SafetyQuote
C26	Admin Controller		+manageUserAccounts() : void +manageDatabaseContents() : void
C27	Risk Assessment Controller		+verifyInformation(submission ID) : void +analyzeRiskData(reportID) : void
C28	Submission Controller		+submitInformation(content) : SubmittedInformation
C29	Route Assistant		+generateRoute(start, end) : Route +calculateAlternativeRoutes() : list<Route>

Table 3.3.1 Class table

3.4 Algorithm and/or Query Design

The following codes detail the core logic of the Sentinel system through its algorithms and SQL queries. The pseudocode algorithms outline the step-by-step operational flow for each specific use case, while the SQL queries provide the Data Definition Language (DDL) statements used to establish the system's underlying data structures and classes.

Main Program

```

PROGRAM TravelSafetyApp
  // Initialization of MVC Components
  INITIALIZE Database_Connection
  INITIALIZE Controllers (Auth, Map, SafetyContent, etc.)
  INITIALIZE Views (Login, Dashboard, MapView, etc.)

  SET isLoggedIn = FALSE
  SET currentUser = NULL

  // 1. Authentication Phase
  WHILE isLoggedIn IS FALSE DO
    LoginView.displayLoginForm()
    INPUT username, password

    // Controller attempts to find User in Model
    SET currentUser = AuthController.login(username, password)

```

```

IF currentUser IS NOT NULL THEN
    SET isLoggedIn = TRUE
    LoginView.showSuccess("Login Successful")
ELSE
    LoginView.showError("Invalid Credentials")
ENDIF
ENDWHILE

// 2. Dashboard Loading Phase
// MainDashboard requests data from SafetyContentController
SET dailyQuote = SafetyContentController.getRandomQuote()
MainDashboard.displayRandomQuote(dailyQuote)
MainDashboard.showWelcomeMessage(currentUser.username)

// 3. Main Application Loop
SET sessionActive = TRUE

WHILE sessionActive IS TRUE DO
    // Render menu based on User role (Traveler, Admin, Journalist, etc.)
    MainDashboard.renderMenu(currentUser.role)
    INPUT userSelection

    SWITCH userSelection
        CASE "MAP_SEARCH":
            CALL Module_MapAndRouting()
        ENDCASE

        CASE "SAFETY_INFO":
            CALL Module_SafetyInformation()
        ENDCASE

        CASE "SUBMIT_INTEL":
            // Only for Journalist or DataAnalyst
            IF currentUser.role IS "Journalist" OR "DataAnalyst" THEN
                CALL Module_Submission()
            ELSE
                PRINT "Access Denied"
            ENDIF
        ENDCASE

        CASE "VERIFY_DATA":
            // Only for DataAnalyst

```

```

        IF currentUser.role IS "DataAnalyst" THEN
            CALL Module_RiskVerification()
        ENDIF
    ENDCASE

    CASE "ADMIN_PANEL":
        // Only for Administrator
        IF currentUser.role IS "Administrator" THEN
            CALL Module_Administration()
        ENDIF
    ENDCASE

    CASE "LOGOUT":
        AuthController.logout()
        SET sessionActive = FALSE
        SET isLoggedIn = FALSE
    ENDCASE
ENDSWITCH
ENDWHILE

PRINT "System Shutdown"
ENDPROGRAM

```

Map and Routing Function

```

FUNCTION Module_MapAndRouting()
    MapView.displayInteractiveMap()
    INPUT searchString

    // Controller searches Model (Destination)
    SET destinationList = MapController.searchDestination(searchString)

    IF destinationList IS EMPTY THEN
        MapView.showError("No destinations found.")
    ELSE
        MapView.showPins(destinationList)

        PRINT "Do you want to generate a route? (Y/N)"
        INPUT routeChoice

        IF routeChoice IS "Y" THEN
            INPUT startLocation
            INPUT endLocation
        ENDIF
    ENDIF
ENDFUNCTION

```

```

    // MapController delegates calculation to RouteAssistant
    // RouteAssistant creates Route object
    SET calculatedRoute = RouteAssistant.generateRoute(startLocation,
endLocation)

    // View renders the Route Model
    MapView.renderRoute(calculatedRoute)
ENDIF
ENDIF
ENDFUNCTION

```

Safety Information Function

```

FUNCTION Module_SafetyInformation()
    INPUT targetCountry

    // Controller fetches various Models
    SET tipsList = SafetyContentController.fetchSafetyTips(targetCountry)
    SET alertList = SafetyContentController.getAlertsForLocation(targetCountry)
    SET advisory = SafetyContentController.getTravelAdvisory(targetCountry)

    // View displays data
    SafetyInfoView.displayTips(tipsList)
    SafetyInfoView.displayAlerts(alertList)
    SafetyInfoView.displayAdvisories(advisory)

    // Feature: Save a Reminder (Traveler specific)
    PRINT "Would you like to save a safety reminder? (Y/N)"
    INPUT saveChoice

    IF saveChoice IS "Y" THEN
        INPUT reminderText
        // Controller saves to SafetyReminder Model associated with User
        SafetyContentController.saveSafetyReminder(currentUser.userID, reminderText)
        PRINT "Reminder Saved."
    ENDIF
ENDFUNCTION

```

Submission and Risk Verification Functions

```

// Used by Journalists to submit on-ground intel
FUNCTION Module_Submission()
    SubmissionForm.renderInputFields()
    INPUT contentData

```

```

// Controller creates SubmittedInformation Model
SET submissionStatus = SubmissionController.submitInformation(contentData)

IF submissionStatus IS TRUE THEN
    SubmissionForm.displaySuccessMessage()
ELSE
    SubmissionForm.showError("Submission Failed")
ENDIF
ENDFUNCTION

// Used by Data Analysts to validate submissions
FUNCTION Module_RiskVerification()
    // Fetch pending items (Pseudocode for DB access)
    SET pendingItems = RiskAssessmentController.getPendingSubmissions()

    FOR EACH item IN pendingItems DO
        DISPLAY item.submittedContent
        INPUT analystDecision // "Approve" or "Reject"

        IF analystDecision IS "Approve" THEN
            // 1. Verify the specific submission
            RiskAssessmentController.verifyInformation(item.submissionID)

            // 2. Aggregate into a formal Risk Report
            SET newReport = NEW RiskReport()
            newReport.riskData = item.submittedContent
            newReport.source = "Verified Submission"

            RiskAssessmentController.analyzeRiskData(newReport.reportID)
            PRINT "Item Verified and Report Created."
        ELSE
            PRINT "Item Rejected."
        ENDIF
    ENDFOR
ENDFUNCTION

```

SQL DDL QUERY

```

-- 1. Base User Table
CREATE TABLE Users (
    userID INT PRIMARY KEY AUTO_INCREMENT,

```

```

username VARCHAR(255) NOT NULL UNIQUE,
password VARCHAR(255) NOT NULL -- Should be stored as a hash
);

-- 2. User Sub-types (Inheritance)
CREATE TABLE Travelers (
    travelerID INT PRIMARY KEY,
    userID INT NOT NULL,
    FOREIGN KEY (userID) REFERENCES Users(userID) ON DELETE CASCADE
);

CREATE TABLE Administrators (
    administratorID INT PRIMARY KEY,
    userID INT NOT NULL,
    FOREIGN KEY (userID) REFERENCES Users(userID) ON DELETE CASCADE
);

CREATE TABLE TravelAgencyPartners (
    partnerID INT PRIMARY KEY,
    userID INT NOT NULL,
    FOREIGN KEY (userID) REFERENCES Users(userID) ON DELETE CASCADE
);

CREATE TABLE DataAnalysts (
    analystID INT PRIMARY KEY,
    userID INT NOT NULL,
    FOREIGN KEY (userID) REFERENCES Users(userID) ON DELETE CASCADE
);

CREATE TABLE Journalists (
    journalistID INT PRIMARY KEY,
    userID INT NOT NULL,
    FOREIGN KEY (userID) REFERENCES Users(userID) ON DELETE CASCADE
);

-- 3. Core Entities
CREATE TABLE Destinations (
    destinationID INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(255) NOT NULL,
    country VARCHAR(100) NOT NULL,
    riskLevel VARCHAR(50)
);

CREATE TABLE RiskReports (
    reportID INT PRIMARY KEY AUTO_INCREMENT,
    riskData TEXT,
    source VARCHAR(255)
);

CREATE TABLE SafetyQuotes (
    quoteID INT PRIMARY KEY AUTO_INCREMENT,

```

```

quoteContent TEXT NOT NULL,
dateAdded DATE DEFAULT (CURRENT_DATE)
);

-- 4. Entities with Relationships (Foreign Keys)
CREATE TABLE SafetyTips (
    tipID INT PRIMARY KEY AUTO_INCREMENT,
    tipType VARCHAR(100),
    content TEXT,
    country VARCHAR(100)
);

CREATE TABLE SafetyReminders (
    reminderID INT PRIMARY KEY AUTO_INCREMENT,
    travelerID INT NOT NULL,
    content TEXT,
    FOREIGN KEY (travelerID) REFERENCES Travelers(travelerID) ON DELETE CASCADE
);

CREATE TABLE SafetyAlerts (
    alertID INT PRIMARY KEY AUTO_INCREMENT,
    destinationID INT NOT NULL,
    content TEXT,
    country VARCHAR(100),
    FOREIGN KEY (destinationID) REFERENCES Destinations(destinationID) ON DELETE
CASCADE
);

CREATE TABLE Routes (
    routeID INT PRIMARY KEY AUTO_INCREMENT,
    destinationID INT, -- Related to 'included in' relationship
    startPoint VARCHAR(255),
    endPoint VARCHAR(255),
    stops JSON, -- Stores the 'list' of stops as a JSON array
    FOREIGN KEY (destinationID) REFERENCES Destinations(destinationID) ON DELETE SET
NULL
);

CREATE TABLE TravelAdvisories (
    advisoryID INT PRIMARY KEY AUTO_INCREMENT,
    destinationID INT NOT NULL,
    type VARCHAR(100),
    content TEXT,
    source VARCHAR(255),
    FOREIGN KEY (destinationID) REFERENCES Destinations(destinationID) ON DELETE
CASCADE
);

CREATE TABLE KeySafetyPoints (
    pointID INT PRIMARY KEY AUTO_INCREMENT,
    destinationID INT NOT NULL,

```

```

content TEXT,
country VARCHAR(100),
FOREIGN KEY (destinationID) REFERENCES Destinations(destinationID) ON DELETE
CASCADE
);

CREATE TABLE SubmittedInformation (
    submissionID INT PRIMARY KEY AUTO_INCREMENT,
    journalistID INT, -- The one who submits
    analystID INT, -- The one who verifies
    reportID INT, -- Aggregated into a Risk Report
    submittedContent TEXT,
    status BOOLEAN DEFAULT FALSE,
    FOREIGN KEY (journalistID) REFERENCES Journalists(journalistID),
    FOREIGN KEY (analystID) REFERENCES DataAnalysts(analystID),
    FOREIGN KEY (reportID) REFERENCES RiskReports(reportID)
);

```

4 Traceability Matrix (Requirement Traceability Matrix)

The Requirement Traceability Matrix in Table 4.1 correlates Functional Requirement (FR) codes with their specific requirement and use case names. For this project, the functional requirement and use case names correspond exactly.

FR Code	Functional Requirement Name	Use Case Name
FR-01	Destination Details Presentation	Search Destination Safety Profile
FR-02	Safety Alerts and Contacts Display	Check Real-time Destination Safety Alerts
FR-03	Key Safety Points Highlighting	Review Key Safety Points
FR-04	Short Safety Tips Display	View Short Safety Tips
FR-05	Personal Reminder Storage	Save Safety Reminder
FR-06	Short Safety Quote Display	View Short Safety Quote
FR-07	Interactive Risk Map Visualization	Explore Interactive Risk Map
FR-08	User Registration System	Register New Account
FR-09	User Account Administration	Manage User Profiles
FR-10	Destination Search Capability	Search for Destinations
FR-11	Information Verification Mechanism	Verify Submitted Data
FR-12	Data Filtering and Sorting Tools	Filter and Sort Results
FR-13	Database Content Management	Administer System Data
FR-14	Route Calculator Engine	Generate Travel Route

Table 4.1 Traceability matrix

Attachment A: Entity Relationship Diagram

This section presents the following Entity Relationship Diagram (ERD) for the system. The ERD illustrates the entities, attributes, and relationships that form the structure of the database.

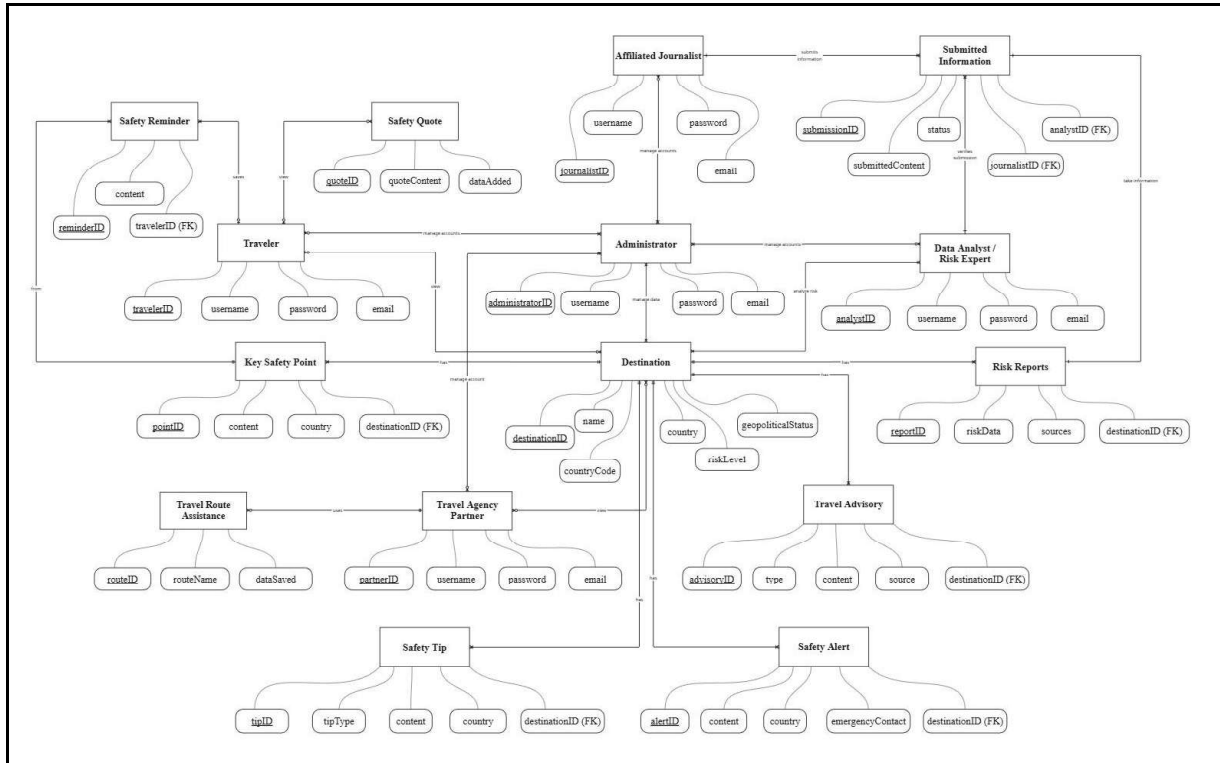


Figure A.1 Sentinel's Entity Relationship Diagram

Attachment B: Relational Schema

This section presents the Relational Schema derived from the ERD. It defines the database tables, their attributes, primary keys, and foreign keys used in the implementation of the system.

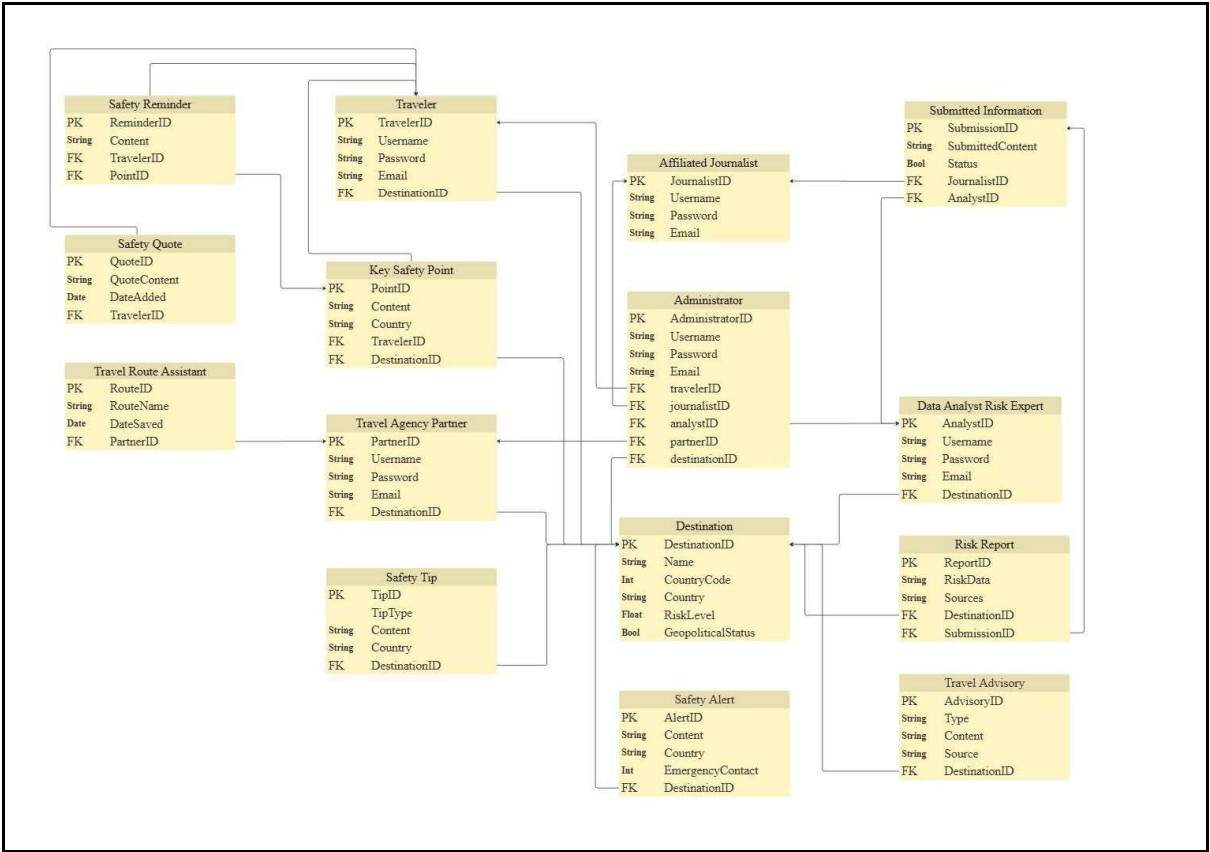


Figure B.1 Sentinel’s Relational Schema

Attachment C: MVC Class Diagram Algorithms

This section presents the Relational Schema derived from the ERD. It defines the database tables, their attributes, primary keys, and foreign keys used in the implementation of the system.

```
---
config:
  theme: mc
  layout: elk
---
classDiagram
direction TB
    namespace Model {
    class User {
        +int userID
        +string username
        #string password
    }

    class Traveler {
        +int travelerID
    }

    class Administrator {
        +int administratorID
    }

    class TravelAgencyPartner {
        +int partnerID
    }

    class DataAnalyst {
        +int analystID
    }

    class Journalist {
        +int journalistID
    }

    class Destination {
        +int destinationID
    }
```

```

        +string name
        +string country
        +string riskLevel
        +getInformation()
    }

class SafetyTip {
    +int tipID
    +string tipType
    +string content
    +string country
    +getTip()
}

class SafetyReminder {
    +int reminderID
    +string content
    +getReminder()
}

class SafetyAlert {
    +int alertID
    +string content
    +string country
    +getAlert()
}

class Route {
    +int routeID
    +string startPoint
    +string endPoint
    +list stops
}

class SubmittedInformation {
    +int submissionID
    +string submittedContent
    +bool status
}

class RiskReport {
    +int reportID

```

```

        +string riskData
        +string source
    }

    class TravelAdvisory {
        +int advisoryID
        +string type
        +string content
        +string source
        +getAdvice()
    }

    class KeySafetyPoint {
        +int pointID
        +string content
        +string country
        +getPoint()
    }

    class SafetyQuote {
        +int quoteID
        +string quoteContent
        +date dateAdded
    }

    }
    namespace View {
    class LoginView {
        +displayLoginForm()
        +displayRegisterForm()
        +showError(msg)
    }

    class MainDashboard {
        +renderMenu()
        +showWelcomeMessage()
        +displayRandomQuote(quote)
    }

    class MapView {
        +displayInteractiveMap()
        +renderRoute(route)
    }

```

```

        +showPins(destinations)
    }

    class SafetyInfoView {
        +displayTips(tips)
        +displayAlerts(alerts)
        +displayAdvisories(advisories)
    }

    class AdminPanel {
        +displayUserList()
        +displayVerificationQueue()
    }

    class SubmissionForm {
        +renderInputFields()
        +displaySuccessMessage()
    }

    }
    namespace Controller {
    class AuthController {
        +login(user, pass)
        +register(userData)
    }

    class MapController {
        +searchDestination(query)
        +generateRoute(start, end)
    }

    class SafetyContentController {
        +fetchSafetyTips()
        +saveSafetyReminder(userId, reminder)
        +getAlertsForLocation(country)
        +getRandomQuote()
    }

    class AdminController {
        +manageUserAccounts()
        +manageDatabaseContents()
    }
}

```

```

class RiskAssessmentController {
    +verifyInformation(submissionID)
    +analyzeRiskData(reportID)
}

```

```

class SubmissionController {
    +submitInformation(content)
}

```

```

class RouteAssistant {
    +generateRoute(start, end)
    +calculateAlternativeRoutes()
}

}

```

User <|-- Traveler

User <|-- Administrator

User <|-- TravelAgencyPartner

User <|-- DataAnalyst

User <|-- Journalist

Destination "1" -- "0..*" SafetyAlert : relates to

Destination "1" -- "1..*" KeySafetyPoint : has

Destination "1" -- "1..*" TravelAdvisory : has

Destination "1" -- "0..*" Route : included in

Traveler "1" -- "0..*" SafetyReminder : saves

DataAnalyst "1" -- "0..*" SubmittedInformation : verifies

Journalist "1" -- "0..*" SubmittedInformation : submits

RiskReport "1" -- "1..*" SubmittedInformation : aggregates

MapController "1" ..> "0..*" Destination : searches

MapController "1" ..> "0..*" Route : creates

MapController "1" ..> "1" RouteAssistant : uses

RouteAssistant "1" ..> "0..*" Route : creates

SafetyContentController "1" ..> "1..*" SafetyTip : fetches

SafetyContentController "1" ..> "0..*" SafetyAlert : fetches

SafetyContentController "1" ..> "1" SafetyQuote : fetches random

SubmissionController "1" ..> "1" SubmittedInformation : creates

RiskAssessmentController "1" ..> "0..*" SubmittedInformation : verifies

LoginView "1" ..> "1" AuthController : login action

MapView "1" ..> "1" MapController : map action

SafetyInfoView "1" ..> "1" SafetyContentController : load info

SubmissionForm "1" ..> "1" SubmissionController : submit action
MainDashboard "1" ..> "1" AuthController : logout
MainDashboard "1" ..> "1" MapController : navigation
MainDashboard "1" ..> "1" SafetyContentController : requests quote

Attachment Z: List of Group Members and Contribution

The following Table Z.1 lists the members of Group Anti-Heroes and outlines their respective contributions to the development of this DPPL document.

#	Full Name	Student ID	Contribution
			<i>Chapter: Description (Exclude FR Resp.)</i>
1.	Farrell [REDACTED] [REDACTED]	[REDACTED]	- FR Responsibility: FR-11, FR-12, FR-15 - 2.1: Contents - 3.2: Class diagram finishing - 3.4: SQL DDL query design - Attachment A: ERD design - Attachment B: Relational Schema design
2.	Putu Argya [REDACTED]	[REDACTED]	- FR Responsibility: FR-13, FR-14 - 1.4: References list - 3.2: Class diagram finishing - 3.4: SQL DDL query design - Attachment A: ERD design - Attachment B: Relational Schema design
3.	Axella [REDACTED] [REDACTED]	[REDACTED]	- FR Responsibility: FR-06, FR-07, FR-08 - List of Changes - Table of Contents, Tables, and Figures - 2.2: Contents - 3.2: Class diagram design - Attachment A: ERD design - Attachment B: Relational Schema design - Attachment Z: Contents
4.	Anita [REDACTED] [REDACTED]	[REDACTED]	- FR Responsibility: FR-04, FR-05 - List of Changed Pages - 1.3: Contents

			- 3.1: Use case realization - 3.3: Class table contents - 4: Traceability matrix table contents - Attachment A: ERD design - Attachment B: Relational Schema design
5.	Chahat		- FR Responsibility: FR-09, FR-10 - 1.1: Contents - Attachment A: ERD thoughts
6.	Naisha		- FR Responsibility: FR-01, FR-02, FR-03 - 1.2: Contents - Attachment A: ERD optimization analysis - Attachment C: MVC algorithm documentation

Table Z.1 Member contribution